THE COMPLETE

CHORDS PRACTICE BOOK

A Structured, Step-by-Step Course for Keyboard

Beginner to Intermediate Level

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Chapter 1 · Introduction to Chords

Every chord you will ever play — from a simple three-note triad to a rich, jazzy extended chord — is built from the same small set of building blocks: notes, intervals, and a starting point called the root. This chapter lays that foundation carefully, because everything later in this book (major, minor, seventh, extended, and altered chords) is really just a variation on the ideas introduced here.

1.1 What Is a Chord?

A chord is three or more notes played together, at the same time, that combine to create a single unit of harmony. Where a scale is notes played one after another (melodically), a chord is notes stacked and sounded together (harmonically).

On the keyboard this is intuitive: if you play C, E, and G at the same moment with your right hand, you have played a C Major chord. Each note keeps its own identity, but together they create a new, combined sound — a "color" — that is different from any single note alone.

> **Key Idea**
> A chord = 3 or more notes + sounded together + built on a systematic pattern of intervals. Two notes played together are called an interval or a "dyad," not a chord.

Why Chords Matter

- Chords provide harmony — the "backing" that supports a melody.
- Chords establish mood: major chords generally sound bright and stable; minor chords sound darker or more emotional.
- Chord progressions (chords played in sequence) are the harmonic skeleton of almost all music you know — pop, film score, devotional, classical, and jazz alike.
- Understanding how chords are built means you can work out ANY chord yourself, in any key, without memorizing hundreds of shapes by rote.

1.2 Notes, Intervals, and Chord Formulas

Before we can build a chord, we need a precise way to measure the distance between two notes. That distance is called an interval, and it is measured in semitones (half-steps) — the smallest distance between two adjacent keys on the keyboard, black or white.

The Semitone: Your Measuring Tape

On a keyboard, moving from any key to the very next key — white to black, black to white, or white to white where there is no black key between (E to F, and B to C) — is one semitone. Two semitones make one whole tone.

Common Intervals Used in Chord Building

Interval Name	Semitones	Example (from C)	Sound Quality
Minor 2nd	1	C → Db	Very tense, dissonant

Interval Name	Semitones	Example (from C)	Sound Quality
Major 2nd	2	C → D	Mildly tense
Minor 3rd	3	C → Eb	Dark, minor color
Major 3rd	4	C → E	Bright, major color
Perfect 4th	5	C → F	Open, suspended
Tritone	6	C → Gb/F#	Unstable, dissonant
Perfect 5th	7	C → G	Stable, open, "power" sound
Minor 6th	8	C → Ab	Rich, wistful
Major 6th	9	C → A	Sweet, warm
Minor 7th	10	C → Bb	Bluesy, unresolved
Major 7th	11	C → B	Dreamy, jazzy
Octave	12	C → C	Same note, higher

The two intervals that matter most when you first start building chords are the Major 3rd (4 semitones) and the Minor 3rd (3 semitones). Almost every basic chord is a stack of thirds — and whether each third is major or minor determines the chord's quality.

The Chord Formula

A chord formula describes a chord using scale-degree numbers (1, 3, 5, 7…) instead of specific note names. This is what makes the formula "movable" — the same formula works in every key.

Chord Type	Formula	Semitone Structure (Root→3rd→5th)	Example (C)
Major Triad	1 – 3 – 5	4 + 3 semitones	C – E – G
Minor Triad	1 – b3 – 5	3 + 4 semitones	C – Eb – G
Diminished Triad	1 – b3 – b5	3 + 3 semitones	C – Eb – Gb
Augmented Triad	1 – 3 – #5	4 + 4 semitones	C – E – G#

> **Practice Tip**
> Memorize the formula, not just the notes. If you know "Major = 1–3–5 = 4 semitones then 3 semitones," you can build a major triad starting on ANY key in seconds — even ones you have never practiced before.

1.3 How Chords Are Built From Scales

Chords do not appear randomly — they are built directly out of scales, using a technique called stacking thirds (also called "tertian harmony"). Here is the process, step by step, using the C Major scale.

Write out the scale: C – D – E – F – G – A – B – (C)

Number each note as a scale degree: 1(C) 2(D) 3(E) 4(F) 5(G) 6(A) 7(B)

To build a triad on any scale degree, take that note, skip one, take the next, skip one, take the next — i.e., take every other note.

For a triad built on degree 1 (C): take 1 (C), 3 (E), 5 (G) → C–E–G → C Major.

The 7 Chords Hiding Inside Every Major Scale

If you repeat this "every other note" process starting on each of the seven scale degrees, you get seven naturally occurring triads — this is called harmonizing the scale.

Degree	Root	Notes	Chord	Quality	Roman Numeral
1	C	C–E–G	C	Major	I
2	D	D–F–A	Dm	Minor	ii
3	E	E–G–B	Em	Minor	iii
4	F	F–A–C	F	Major	IV
5	G	G–B–D	G	Major	V
6	A	A–C–E	Am	Minor	vi
7	B	B–D–F	Bdim	Diminished	vii°

> **Why This Matters**
>
> This pattern — Major, minor, minor, Major, Major, minor, diminished (I ii iii IV V vi vii°) — is IDENTICAL in every major key. Once you memorize this pattern once, you know the "home chords" of all twelve major keys.

1.4 The Root Note

The root note is the note a chord is named after and built upon — the foundation stone of the chord. In a C Major chord (C–E–G), C is the root. Everything else in the chord is measured as a distance (interval) upward from the root.

Root Position vs. Inversions

When the root is the lowest-sounding note of the chord, the chord is in root position. If a different chord tone is placed at the bottom instead, the chord is inverted — it still has the same name and the same root, only the physical arrangement on the keyboard has changed.

Position	Notes Bottom → Top	Bass Note
Root Position	C – E – G	C (the root)
1st Inversion	E – G – C	E (the 3rd)
2nd Inversion	G – C – E	G (the 5th)

Even in an inversion, the root of the chord does not change — a C chord is still a C chord whether C, E, or G is at the bottom. This idea becomes important later in the book when you learn smooth voice-leading between chords.

> **Common Confusion**
> "Root" and "bass note" are not always the same thing! The root is the note the chord is theoretically built from and named after. The bass note is simply whatever note is physically lowest at a given moment — which changes with inversions.

1.5 Triads vs. Extended Chords

A triad is the simplest possible chord: three notes, stacked in thirds — root, 3rd, and 5th. Every chord family in this book (major, minor, diminished, augmented) starts life as a triad.

An extended chord adds one or more additional notes above the triad, continuing the same "stack of thirds" logic — a 7th, then a 9th, then an 11th, then a 13th. Each added note layers in more color, tension, and sophistication.

From Triad to Extended: The Same Stack, Taller

Chord	Notes (from C)	Tones Stacked	Character
Triad	C – E – G	1–3–5	Simple, stable, clean
7th Chord	C – E – G – Bb	1–3–5–7	Bluesy / jazzy tension
9th Chord	C – E – G – Bb – D	1–3–5–7–9	Lush, colorful
11th Chord	C – E – G – Bb – D – F	1–3–5–7–9–11	Dense, modern/jazz
13th Chord	C – E – G – Bb – D – F – A	1–3–5–7–9–11–13	Full harmonic spectrum

> **Looking Ahead**
> This book will dedicate full chapters later to triads (Major, Minor, Diminished, Augmented) and to extended/7th chords in depth. For now, the goal is simply to recognize: a triad is the root chord shape, and every extended chord is that same triad with more thirds stacked on top.

Chapter Summary

- A chord is 3 or more notes sounded together, built from a systematic stack of intervals.
- Intervals are measured in semitones; Major and minor 3rds are the building blocks of basic chords.
- A chord formula (e.g., 1–3–5) describes a chord in movable, key-independent terms.
- Chords are built by stacking every other note of a scale ("tertian harmony"), which produces 7 naturally occurring chords per major scale.
- The root is the note a chord is named after and measured from — it stays the same even when the chord is inverted.
- Triads (root–3rd–5th) are the foundation; extended chords add further thirds (7th, 9th, 11th, 13th) on top of the same triad.

Practice Exercises

Work through these in order. Use a real keyboard, not just your eyes — the goal is for your hands and ears to recognize these patterns, not only your brain.

Exercise 1: Semitone Counting

- On your keyboard, starting from Middle C, count and say aloud each semitone up to the C one octave higher (there are 12).
- Now do the same starting from F, then from A.
- Play a Major 3rd (4 semitones) and a minor 3rd (3 semitones) starting from 5 different white keys: C, D, F, G, A.

Self-check:

☐ I can find any Major 3rd or minor 3rd from a random starting note within 5 seconds.

☐ I can explain, out loud, the difference in semitone count between a Major and minor 3rd.

Exercise 2: Building Triads by Formula

- Using the formula 1–3–5, build (play) a Major triad starting on each of these roots: C, F, G, D, A.
- Using the formula 1–b3–5, build a minor triad on the same five roots.
- For each triad, say the letter names of all three notes out loud as you play them.

Self-check:

☐ I built each triad without looking up the notes — using only the semitone formula.

☐ I can name all three notes in each triad immediately after playing it.

Exercise 3: Harmonizing the C Major Scale

- Play the C Major scale, one octave, hands separate, slowly and evenly.
- Now build and play the triad on every scale degree (I through vii°) exactly as shown in the table in section 1.3.
- Say the chord quality out loud for each one as you play it: "Major, minor, minor, Major, Major, minor, diminished."

Self-check:

☐ I can play all 7 triads of the C Major scale in order without stopping.

☐ I have memorized the quality pattern: Major–minor–minor–Major–Major–minor–diminished.

Exercise 4: Root Position and Inversions

- Play a C Major triad in root position (C–E–G).
- Now play its 1st inversion (E–G–C) and 2nd inversion (G–C–E).
- Play all three positions back-to-back, slowly, and notice that the chord "name" never changes even though the shape does.
- Repeat this whole exercise for G Major and F Major triads.

Self-check:

☐ I can play all three positions of C, F, and G Major smoothly.

☐ I can explain the difference between the "root" and the "bass note" in an inversion.

Exercise 5: Ear and Recognition Check

- Close your eyes (or look away from the keys). Have someone play — or record yourself playing — a Major triad and a minor triad, several times in random order.
- Try to identify by ear alone which one is Major and which is minor.
- Then try the same test comparing a triad to a 7th chord, to start training your ear for "added color."

Self-check:

☐ I correctly identify Major vs minor by ear at least 8 times out of 10.

☐ I can describe, in my own words, how a 7th chord "feels" different from a plain triad.

Next Chapter Preview

Chapter 2 will take the Major triad formula from this chapter and apply it systematically across all 12 keys, building full command of Major chords on the keyboard.

Chapter 2 · Major Scale → Chords

In Chapter 1 you saw a preview of how chords come out of a scale. In this chapter we slow that process down, make it completely systematic, and extend it to all twelve major keys — so that "harmonizing a scale" becomes a skill you own everywhere on the keyboard, not just in the key of C.

2.1 Major Scale Review

The major scale is built from a fixed pattern of whole steps (W) and half steps (H), starting on the root note: W – W – H – W – W – W – H. This pattern is what gives every major scale, in every key, the same bright, "happy" sound.

Degree	1	2	3	4	5	6	7	8 (octave)
C Major	C	D	E	F	G	A	B	C
G Major	G	A	B	C	D	E	F#	G
F Major	F	G	A	Bb	C	D	E	F
D Major	D	E	F#	G	A	B	C#	D

Step interval pattern between each degree: C→D (W), D→E (W), E→F (H), F→G (W), G→A (W), A→B (W), B→C (H). Notice the half-steps always fall between degrees 3–4 and 7–8 — this never changes, in any key.

> **Quick Refresher**
> If you can spell any major scale using the W-W-H-W-W-W-H formula, you already have everything you need to build chords in that key. Scale spelling is the prerequisite skill for this whole chapter.

2.2 Building Chords From the Major Scale

The technique is the same one introduced in Chapter 1: stack thirds by taking every other note of the scale. To build a triad on any scale degree, take that note, skip one, take the next, skip one, take the next.

1. Write the scale across an octave and a half so you have room to "skip ahead" past degree 7 into the next octave (e.g., C D E F G A B C D E for C Major).
2. Pick a starting scale degree — this becomes the root of your new chord.
3. Skip one note, take the next — this becomes the 3rd of your chord.
4. Skip one more note, take the next — this becomes the 5th of your chord.
5. Check the quality by measuring the semitone gaps: 4+3 semitones = Major, 3+4 = minor, 3+3 = diminished.

Example: starting on degree 5 of C Major (G), skip A, take B (3rd), skip C, take D (5th) → G–B–D → G Major triad.

2.3 The 1st, 3rd, and 5th Notes

Every triad — no matter which scale degree it is built from — is described using the same three functional labels relative to its OWN root, not the root of the parent scale:

- 1st (Root): the note the chord is built from and named after; the harmonic "anchor."
- 3rd: the note that determines whether the chord sounds Major or minor — this is the single most important note for chord quality.
- 5th: the note that reinforces and stabilizes the chord; in most basic triads it is a Perfect 5th (7 semitones) above the root.

> **Key Idea**
> The "1–3–5" labels always reset relative to each chord's own root. When you build a triad on scale degree 2 (D, in C Major), that triad's own 1st is D, its own 3rd is F, and its own 5th is A — even though D, F, and A are degrees 2, 4, and 6 of the parent C Major scale.

Worked Example: Every Triad's Own 1–3–5 (Key of C)

Scale Degree	Chord	Chord's Own Root (1)	Chord's Own 3rd	Chord's Own 5th
I	C	C	E	G
ii	Dm	D	F	A
iii	Em	E	G	B
IV	F	F	A	C
V	G	G	B	D
vi	Am	A	C	E
vii°	Bdim	B	D	F

2.4 Diatonic Chords

A diatonic chord is any chord built entirely from the notes of one particular scale — no notes borrowed from outside that key. The seven chords you built in section 2.2 (and in Chapter 1) are the diatonic chords of C Major.

Diatonic chords matter because they are the chords that "belong" naturally together — they share a key signature and, when combined into a progression, they automatically sound cohesive without needing any special resolution.

The Diatonic Quality Pattern (Every Major Key)

Degree	1	2	3	4	5	6	7
Quality	Major	minor	minor	Major	Major	minor	diminished

> **Key Idea**

This Major–minor–minor–Major–Major–minor–diminished pattern is fixed. It is the single most useful fact in this chapter: memorize it once, and you instantly know the quality of every diatonic chord in every major key, without recalculating semitones each time.

2.5 The Roman Numeral System

Roman numerals are a shorthand for describing diatonic chords by their scale degree instead of their letter name. This is what lets musicians talk about a chord progression in a way that works in any key.

- UPPERCASE numeral (I, IV, V) = Major chord
- lowercase numeral (ii, iii, vi) = minor chord
- lowercase numeral with a small circle (vii°) = diminished chord

Roman Numeral	Degree	Quality	In Key of C	In Key of G
I	1	Major	C	G
ii	2	minor	Dm	Am
iii	3	minor	Em	Bm
IV	4	Major	F	C
V	5	Major	G	D
vi	6	minor	Am	Em
vii°	7	diminished	Bdim	F#dim

This is why a progression like "I–V–vi–IV" is instantly transposable: in C Major it is C–G–Am–F; in G Major it is G–D–Em–C. Same numerals, same emotional shape, different letters.

Practice Tip

When learning a new song by ear, try to identify its progression in Roman numerals rather than letter names. You will then be able to play that same progression in ANY key a singer asks for, instantly.

2.6 Chords in All Major Keys

Because the diatonic quality pattern (Major-minor-minor-Major-Major-minor-diminished) never changes, building the seven chords of any major key is simply a matter of (1) spelling that key's scale correctly, and (2) applying the pattern in order.

Reference Table: Diatonic Chords in All 12 Major Keys

Key	I	ii	iii	IV	V	vi	vii°
C	C	Dm	Em	F	G	Am	Bdim
G	G	Am	Bm	C	D	Em	F#dim

Key	I	ii	iii	IV	V	vi	vii°
D	D	Em	F#m	G	A	Bm	C#dim
A	A	Bm	C#m	D	E	F#m	G#dim
E	E	F#m	G#m	A	B	C#m	D#dim
B	B	C#m	D#m	E	F#	G#m	A#dim
F#	F#	G#m	A#m	B	C#	D#m	E#dim
F	F	Gm	Am	Bb	C	Dm	Edim
Bb	Bb	Cm	Dm	Eb	F	Gm	Adim
Eb	Eb	Fm	Gm	Ab	Bb	Cm	Ddim
Ab	Ab	Bbm	Cm	Db	Eb	Fm	Gdim
Db	Db	Ebm	Fm	Gb	Ab	Bbm	Cdim

How to Use This Table

Do not simply memorize this table by rote. Use it to check your own work after you build each key's chords yourself using the scale + pattern method — the table is a reference and an answer key, not a shortcut around the exercises below.

Chapter Summary

- The major scale follows a fixed W-W-H-W-W-W-H step pattern in every key.
- Chords are built by stacking every other scale note (skip-one, take-one) starting on any degree.
- Every triad has its own 1st (root), 3rd, and 5th, measured from ITS OWN root — not the parent scale's root.
- A diatonic chord is any chord built entirely from one scale's notes; every major scale produces 7 diatonic chords.
- The diatonic quality pattern — Major, minor, minor, Major, Major, minor, diminished (I ii iii IV V vi vii°) — is identical in all 12 major keys.
- Roman numerals (I, ii, iii, IV, V, vi, vii°) describe chords by scale degree and quality, making progressions instantly transposable between keys.

Practice Exercises

These exercises move from single-key mastery (C Major) toward flexible, all-key fluency. Do not skip ahead until each step feels comfortable.

Exercise 1: Spelling Major Scales

- Using the W-W-H-W-W-W-H formula, spell out (write and then play) the G Major, D Major, and F Major scales from memory.
- Check each one against the table in section 2.1 only after attempting it yourself.
- Play each scale hands separately, ascending and descending, at a slow, even tempo.

Self-check:

☐ I can spell G, D, and F Major scales without looking at a reference.

☐ I can play all three scales smoothly, ascending and descending.

Exercise 2: Harmonizing a New Key by Hand

- Choose a key you have not yet worked with in this book (e.g., D Major or F Major).
- Using the skip-one-take-one method, build all seven diatonic triads in that key on paper first, then on the keyboard.
- Label each chord with its Roman numeral and its quality (Major/minor/diminished) as you go.
- Check your results against the reference table in section 2.6.

Self-check:

☐ I built all 7 chords correctly without directly copying the table first.

☐ I labeled each chord's Roman numeral correctly on the first attempt.

Exercise 3: Own Root, 3rd, and 5th Drill

- Play the ii chord (Dm) in C Major. Say out loud its OWN root, 3rd, and 5th (D, F, A) — not C Major's scale degree numbers.
- Repeat for the iii, IV, V, and vi chords of C Major.
- Repeat the entire drill in G Major.

Self-check:

☐ I can state a chord's own 1-3-5 immediately after playing it, in two different keys.

☐ I no longer confuse a chord's own 3rd with the parent scale's 3rd degree.

Exercise 4: Roman Numeral Transposition

- Play the progression I – V – vi – IV in the key of C (C – G – Am – F).
- Without writing anything down, transpose and play the same I–V–vi–IV progression in G Major, then in D Major.
- Try one more progression of your choice, e.g., vi – IV – I – V, in three different keys.

Self-check:

☐ I transposed I-V-vi-IV correctly into G and D Major by ear/logic, not by copying the table.

☐ I can explain why the "shape" of the progression stays the same across keys.

Exercise 5: All-Keys Speed Recall

- Pick any major key at random (use a die, a shuffled deck of key names, or ask someone to call one out).
- As fast as you can, name (and then play) that key's I, IV, and V chords — the three most commonly used chords in any key.
- Repeat for at least 6 different random keys, gradually increasing your speed.

Self-check:

☐ I can name the I, IV, and V chords of a random major key within 10 seconds.

☐ I can play the I, IV, and V chords of at least 6 different keys from memory.

Next Chapter Preview

Chapter 3 will move from the Major scale into the Minor scale and its own diatonic chord family, introducing the natural, harmonic, and melodic minor variations.

Chapter 3 · Major Triads

You have already met the Major triad twice — once as the "1–3–5" formula in Chapter 1, and again as the I, IV, and V chords that appear in every major key in Chapter 2. This chapter takes the Major triad out of theory and puts it fully under your hands: building it instantly from any root, playing it correctly in root position, drilling it across all 12 keys, and — just as important — training your ear to recognize it by sound alone.

3.1 The Major Chord Formula: 1–3–5

Every Major triad, in every key, is built from exactly the same formula: Root (1) – Major 3rd (3) – Perfect 5th (5). Measured in semitones from the root, this is 4 semitones up to the 3rd, then 3 more semitones up to the 5th (4 + 3).

Scale Degree	Interval From Root	Semitones From Root	Role in the Chord
1	Unison	0	Root — the anchor and name of the chord
3	Major 3rd	4	Determines the "Major" quality — the bright color
5	Perfect 5th	7	Stabilizes and fills out the chord

> **Key Idea**
> A Major triad is simply "4 semitones, then 3 semitones" measured upward from any starting note. This formula never changes — it is exactly as true on F# as it is on C.

3.2 Building Major Triads

There are two reliable ways to build a Major triad on any root. Strong players can do both, and use whichever is faster in the moment.

Method 1: Semitone Counting

1. Play the root note.
2. Count up exactly 4 semitones (half-steps) — this is the Major 3rd.
3. From the 3rd, count up exactly 3 more semitones — this is the Perfect 5th.
4. Play all three notes together: Root, 3rd, 5th.

Method 2: Major Scale Degrees

1. Spell the Major scale of your chosen root (e.g., for an E Major triad, spell the E Major scale).
2. Take scale degree 1 as the root.
3. Take scale degree 3 as the chord's 3rd.
4. Take scale degree 5 as the chord's 5th.

Example using both methods on E: E Major scale is E–F#–G#–A–B–C#–D#–E, so degrees 1–3–5 give E–G#–B. Checking by semitones: E to G# is 4 semitones, G# to B is 3 semitones. Both methods agree.

Worked Examples Across Different Root Types

Root	Root Type	+4 Semitones (3rd)	+3 More (5th)	Resulting Triad
C	White key	E	G	C – E – G
D	White key	F#	A	D – F# – A
Eb	Black key	G	Bb	Eb – G – Bb
F#	Black key	A#(Bb)	C#	F# – A# – C#

> **Common Mistake**
> Do not simply count "every other white key" — that only works reliably in C Major. On black-key roots, or on white-key roots that need a sharp/flat 3rd (like D, E, A, and B Major), you must count actual semitones or use the Major scale method, not just skip visually along the white keys.

3.3 Root Position

Root position means the chord is voiced with its root as the lowest-sounding note, followed by the 3rd, then the 5th, stacked upward. This is the "default," most fundamental way to play any triad, and the shape you should be able to produce instantly before moving on to inversions.

Playing Root Position on the Keyboard

- Right hand: fingers 1 (thumb) – 3 (middle) – 5 (little finger) map naturally onto Root – 3rd – 5th.
- Left hand: fingers 5 (little finger) – 3 (middle) – 1 (thumb) map onto Root – 3rd – 5th, reading bottom to top.
- Keep a relaxed, slightly curved hand shape — the "1-3-5" stretch on white keys is comfortable for most hand sizes; black-key chords may need a small hand adjustment.

> **Practice Tip**
> Always say the chord's letter name AND its three note names out loud as you play root position — e.g., "C Major: C, E, G." This builds the mental link between the sound, the shape, and the note names simultaneously.

3.4 Practice in All 12 Keys

Fluency means you can produce any Major triad on demand, in any key, without hesitating to calculate it first. The reference table below lists all 12 Major triads in root position — use it to check your work, not to skip the calculation practice in the exercises.

Root	Major Triad (1–3–5)	Root	Major Triad (1–3–5)
C	C – E – G	F#/Gb	F# – A# – C#

Root	Major Triad (1–3–5)	Root	Major Triad (1–3–5)
C#/Db	Db – F – Ab	G	G – B – D
D	D – F# – A	G#/Ab	Ab – C – Eb
D#/Eb	Eb – G – Bb	A	A – C# – E
E	E – G# – B	A#/Bb	Bb – D – F
F	F – A – C	B	B – D# – F#

> **Practice Sequencing**
> Work through the 12 keys in the "circle of fifths" order (C, G, D, A, E, B, F#, Db, Ab, Eb, Bb, F) rather than chromatically. Each new key shares most of its notes with the previous one, which makes the pattern of "what changes" much easier to notice and internalize.

3.5 Ear Training Exercises

Being able to build a Major triad on paper is only half the skill — a well-rounded player can also recognize a Major triad purely by listening. The Major triad has a bright, stable, "resolved" sound; training your ear to catch that quality quickly will pay off every time you learn a song by ear later in this book.

What to Listen For

- Brightness / openness — Major triads tend to sound cheerful, settled, or "complete."
- The 3rd is the key — mentally isolate the middle note; a wide, bright-sounding gap between root and 3rd signals Major.
- Compare, don't just guess — always compare a chord against a clearly Major reference (like C Major) if you are unsure.

Chapter Summary

- Every Major triad follows the fixed formula 1–3–5, or 4 semitones then 3 semitones from the root.
- Major triads can be built either by counting semitones directly or by taking degrees 1, 3, and 5 of the corresponding Major scale.
- Root position places the root as the lowest note, with the 3rd and 5th stacked above it — the default, foundational voicing.
- Fluency requires practicing all 12 Major triads, ideally in circle-of-fifths order, until each can be produced instantly.
- Ear training closes the loop: recognizing a Major triad's bright, stable quality by sound is as important as building it by hand.

Practice Exercises

Work through these exercises in order, and do not rush past Exercise 1 — solid semitone counting is what makes every later exercise fast and accurate.

Exercise 1: Semitone-Count Build

- Starting on C, build a Major triad using ONLY semitone counting (4 semitones, then 3 semitones) — no scale reference allowed.
- Repeat for D, E, F#, and Ab.
- Time yourself — aim to build each triad correctly within 10 seconds.

Self-check:

☐ I built all 5 triads correctly using semitone counting alone.

☐ My average build time is under 10 seconds per triad.

Exercise 2: Root Position Hand Shapes

- Play root position C, F, and G Major triads with the right hand, using fingers 1-3-5, moving smoothly from one to the next.
- Repeat the same three chords with the left hand, using fingers 5-3-1.
- Now play all three chords with both hands together, in root position, in the sequence C → F → G → C.

Self-check:

☐ I can play C, F, and G root-position triads with correct fingering in both hands.

☐ My hand transitions between the three chords are smooth, without looking down.

Exercise 3: All 12 Keys — Circle of Fifths Drill

- Using the circle-of-fifths order (C, G, D, A, E, B, F#, Db, Ab, Eb, Bb, F), build and play each Major triad in root position, one after another.
- Say each triad's three note names out loud as you play it.
- Once comfortable, repeat the full circle without stopping and without checking the reference table.

Self-check:

☐ I completed all 12 keys in circle-of-fifths order without consulting the table.

☐ I can name all three notes of at least 8 of the 12 triads from memory.

Exercise 4: Black-Key Root Focus

- Isolate the four "trickiest" black-key-root triads: Db, Eb, F#, and Ab Major.
- Build each one using the Major-scale-degree method (not semitone counting) as a cross-check.
- Play each 5 times in a row until the hand shape feels automatic.

Self-check:

☐ I can build Db, Eb, F#, and Ab Major triads confidently using either method.

☐ My hand no longer hesitates or fumbles on these black-key shapes.

Exercise 5: Ear Training: Major Recognition

- Have someone else (or a recording/app) play 10 different triads in random order and random keys, without telling you which is which.
- For each one, simply answer: "Is this Major?" — yes or no.
- Afterward, check your answers and note any triads that fooled you; play those specific ones again slowly to hear exactly why.

Self-check:

☐ I correctly identified Major triads at least 8 times out of 10.

☐ I can describe, in my own words, what makes a triad sound "Major" versus something else.

Next Chapter Preview

Chapter 4 will apply the same complete treatment — formula, building method, root position, all-12-keys practice, and ear training — to the Minor triad, so you can directly compare and contrast the two most fundamental chord qualities.

Chapter 4 · Minor Triads

The Minor triad is the Major triad's closest relative — built from the very same root and 5th, changed by moving just one note. That single semitone shift is responsible for one of the most powerful contrasts in all of music: bright versus dark, resolved versus wistful. This chapter builds the Minor triad from the ground up and, just as important, trains you to hear and feel the difference between Major and minor instantly.

4.1 The Minor Chord Formula: 1–♭3–5

A Minor triad is built from Root (1) – minor 3rd (♭3) – Perfect 5th (5). In semitones, that is 3 semitones up to the ♭3rd, then 4 more semitones up to the 5th (3 + 4) — the exact reverse order of the Major triad's 4 + 3.

Scale Degree	Interval From Root	Semitones From Root	Role in the Chord
1	Unison	0	Root — the anchor and name of the chord
♭3	minor 3rd	3	Determines the "minor" quality — the darker color
5	Perfect 5th	7	Stabilizes the chord (same as in Major)

> **Key Idea**
> A minor triad is "3 semitones, then 4 semitones" from the root. Compare this to the Major triad's "4 semitones, then 3 semitones" — the two intervals are simply swapped in order.

4.2 Major vs. Minor Sound

Since a Major and minor triad sharing the same root differ by only one note — the 3rd — that single note carries the entire emotional weight of the chord's quality.

	Major Triad (from C)	Minor Triad (from C)
Notes	C – E – G	C – Eb – G
3rd interval	Major 3rd (4 semitones)	minor 3rd (3 semitones)
Typical character	Bright, happy, stable, resolved	Dark, sad, tender, unresolved
What moved	—	3rd lowered by 1 semitone (E → Eb)

This is why moving between a Major and minor version of the same chord is often called "changing the mode" of the chord — the root and 5th act as a stable frame, while the 3rd swings the emotional color between bright and dark.

> **Practice Tip**
> Play a C Major triad, then lower only the middle note (E to Eb) to hear it become C minor. Do this slowly

and listen closely — this single-note change is the clearest possible demonstration of what "major" and "minor" actually mean.

4.3 Building Minor Triads

As with Major triads, there are two dependable ways to build any Minor triad.

Method 1: Semitone Counting

1. Play the root note.
2. Count up exactly 3 semitones — this is the minor 3rd (♭3).
3. From the ♭3rd, count up exactly 4 more semitones — this is the Perfect 5th.
4. Play all three notes together: Root, ♭3rd, 5th.

Method 2: From the Major Triad

1. Build the Major triad on the same root (using Chapter 3's method).
2. Lower only the 3rd by one semitone; leave the root and 5th untouched.
3. The result is the Minor triad on that root.

Worked Examples

Root	Major Triad	Lower the 3rd	Minor Triad
C	C – E – G	E → Eb	C – Eb – G
D	D – F# – A	F# → F	D – F – A
G	G – B – D	B → Bb	G – Bb – D
F#	F# – A# – C#	A# → A	F# – A – C#

Common Mistake

Only the 3rd changes between a Major and minor triad on the same root — the 5th stays exactly the same. A frequent beginner error is accidentally lowering the 5th as well, which produces a diminished triad instead of a minor one.

4.4 Minor Triads in All 12 Keys

Use this reference table to check your own work after building each Minor triad yourself, the same way you practiced Major triads in Chapter 3.

Root	Minor Triad (1–♭3–5)	Root	Minor Triad (1–♭3–5)
C	C – Eb – G	F#/Gb	F# – A – C#
C#/Db	Db – E – Ab	G	G – Bb – D
D	D – F – A	G#/Ab	Ab – B – Eb

Root	Minor Triad (1–♭3–5)	Root	Minor Triad (1–♭3–5)
D#/Eb	Eb – Gb – Bb	A	A – C – E
E	E – G – B	A#/Bb	Bb – Db – F
F	F – Ab – C	B	B – D – F#

Practice Sequencing

As with Major triads, work through all 12 Minor triads in circle-of-fifths order (C, G, D, A, E, B, F#, Db, Ab, Eb, Bb, F). Because you already know each key's Major triad from Chapter 3, the fastest route is "build Major, then lower the 3rd."

4.5 Major → Minor Comparison Exercises

The exercises below are designed to build fast, reliable switching between Major and minor on the same root, both by hand and by ear — the skill that underlies almost every real chord progression you will play.

Chapter Summary

- A Minor triad follows the formula 1–♭3–5, or 3 semitones then 4 semitones from the root.
- Major and minor triads on the same root share their root and 5th; only the 3rd differs, by a single semitone.
- The fastest way to build a Minor triad is often to build the Major triad first, then lower the 3rd by one semitone.
- Lowering the 5th as well (instead of just the 3rd) is a common mistake — that produces a diminished triad, not a minor one.
- Fluency in all 12 Minor triads, combined with fast Major-to-minor switching, builds the foundation for real-world chord progressions.

Practice Exercises

These exercises pair directly with Chapter 3 — work through them side by side with your Major triad drills for the fastest progress.

Exercise 1: Semitone-Count Build (Minor)

- Starting on C, build a Minor triad using ONLY semitone counting (3 semitones, then 4 semitones).
- Repeat for D, E, F#, and Ab.
- Time yourself — aim for under 10 seconds per triad.

Self-check:

☐ I built all 5 minor triads correctly using semitone counting alone.

☐ My average build time is under 10 seconds per triad.

Exercise 2: Major-to-Minor Conversion Drill

- Play a Major triad on C, then convert it to minor by lowering only the 3rd — say the note names out loud both times ("C E G… C Eb G").

- Repeat the same Major-to-minor conversion for D, G, and F# triads.
- Now do it in reverse: start on the Minor triad and raise the ♭3rd back to build the Major triad.

Self-check:

☐ I converted all 4 Major triads to minor correctly by ear and by hand.

☐ I can do the conversion in both directions (Major→minor and minor→Major).

Exercise 3: All 12 Keys — Minor Circle of Fifths

- Using circle-of-fifths order, build and play each Minor triad in root position, one after another.
- Say each triad's three note names out loud as you play it.
- Repeat the full circle without stopping and without checking the reference table.

Self-check:

☐ I completed all 12 minor keys in circle-of-fifths order without consulting the table.

☐ I can name all three notes of at least 8 of the 12 minor triads from memory.

Exercise 4: Alternating Major/Minor Same-Root Drill

- Pick 5 random roots. For each one, play the Major triad, then immediately the Minor triad, then back to Major.
- Focus on keeping the root and 5th perfectly still — only the middle finger/note should move.
- Increase your speed slightly on each repetition while keeping the pitches accurate.

Self-check:

☐ I can alternate Major/minor on 5 different roots without hesitating.

☐ My root and 5th genuinely stay still — only the 3rd moves.

Exercise 5: Ear Training: Major vs. Minor

- Have someone else (or a recording/app) play 10 triads in random order — a mix of Major and minor, in random keys.
- For each one, answer simply: "Major or minor?"
- Afterward, review any triads that fooled you, and play those specific roots slowly, Major then minor, to re-anchor the sound in your ear.

Self-check:

☐ I correctly identified Major vs. minor at least 8 times out of 10.

☐ I can describe, in my own words, the emotional difference I hear between the two qualities.

Next Chapter Preview

Chapter 5 will complete the triad family by covering Diminished and Augmented triads — the two remaining, less common triad qualities that add tension and color to progressions.

Chapter 5 · Diminished & Augmented Chords

Major and minor triads are the two "stable" chord qualities — each note sits comfortably, and the chord feels at rest. Diminished and Augmented triads are the opposite: both are built from symmetrical, unstable interval stacks that create tension, asking to move somewhere else. This chapter completes the triad family and shows you where these two colorful, less common chords actually appear in real music.

5.1 The Diminished Formula: 1–♭3–♭5

A Diminished triad is built from Root (1) – minor 3rd (♭3) – diminished 5th (♭5). In semitones: 3 semitones up to the ♭3rd, then 3 more semitones up to the ♭5th (3 + 3) — a minor 3rd stacked on top of another minor 3rd.

Scale Degree	Interval From Root	Semitones From Root	Role in the Chord
1	Unison	0	Root — the anchor and name of the chord
♭3	minor 3rd	3	Same as a minor triad's 3rd
♭5	diminished 5th (tritone)	6	Lowered from a Perfect 5th — creates instability

Example: C diminished = C – Eb – Gb. Compare this to C minor (C – Eb – G): the root and ♭3rd are identical, but the 5th is lowered by one more semitone, from G to Gb.

Key Idea

A diminished triad is "3 semitones, then 3 semitones" — two stacked minor 3rds. This perfectly symmetrical structure is what gives it such a tense, unresolved, "searching" sound.

5.2 The Augmented Formula: 1–3–♯5

An Augmented triad is built from Root (1) – Major 3rd (3) – augmented 5th (♯5). In semitones: 4 semitones up to the 3rd, then 4 more semitones up to the ♯5th (4 + 4) — a Major 3rd stacked on top of another Major 3rd.

Scale Degree	Interval From Root	Semitones From Root	Role in the Chord
1	Unison	0	Root — the anchor and name of the chord
3	Major 3rd	4	Same as a Major triad's 3rd
♯5	augmented 5th	8	Raised from a Perfect 5th — creates a floating, unresolved sound

Example: C augmented = C – E – G#. Compare this to C Major (C – E – G): the root and 3rd are identical, but the 5th is raised by one semitone, from G to G#.

Key Idea

An augmented triad is "4 semitones, then 4 semitones" — two stacked Major 3rds, also perfectly symmetrical. This is why an augmented triad can feel like it has no clear "home" — every note is equally spaced from its neighbors.

All Four Triad Qualities Side by Side (Root: C)

Quality	Formula	Semitone Stack	Notes	Character
Major	1–3–5	4 + 3	C–E–G	Bright, stable
minor	1–♭3–5	3 + 4	C–Eb–G	Dark, stable
Diminished	1–♭3–♭5	3 + 3	C–Eb–Gb	Tense, unstable
Augmented	1–3–♯5	4 + 4	C–E–G#	Floating, unstable

5.3 When and Where They Occur

Diminished Triads: The Natural vii°

Recall from Chapter 2 that every major scale naturally produces one diminished triad — on scale degree 7 (vii°). In C Major, this is B diminished (B–D–F). Because it is the only naturally-occurring diminished triad in the key, it has a distinctive job: it creates strong tension that wants to resolve back to the I chord.

- Passing / leading-tone chord: the vii° triad almost always moves to I, acting like a substitute or extension of the V chord's pull toward "home."
- Diminished 7th chords (a 4-note extension, covered in a later chapter) are common in classical and film-score writing for dramatic, suspenseful moments.
- Fully diminished chords are also used as "passing chords" — brief connective chords between two more stable chords, especially in jazz and gospel-influenced styles.

Augmented Triads: The Chromatic Guest

Unlike the diminished triad, the augmented triad is NOT one of the seven naturally occurring diatonic chords in a major or natural minor scale — it must be borrowed or created chromatically (by deliberately altering a note outside the key).

- Dominant augmented: raising the 5th of a V chord (e.g., G to G#) creates tension that pulls even more strongly toward the I chord, common in jazz and gospel cadences.
- Whole-tone / impressionistic color: composers like Debussy and many film composers use augmented triads for a dreamlike, ambiguous, "suspended in air" quality.
- Transitional / passing chord: an augmented triad can smoothly connect two chords a whole step apart, since it shares notes with both.

Key Idea

Diminished triads occur naturally within every major key (as vii°); augmented triads do not occur

naturally and must be deliberately introduced for color or tension. This is why diminished chords feel more "at home" in traditional harmony, while augmented chords feel more exotic.

Diminished Triads in All 12 Keys

Root	Diminished Triad (1–♭3–♭5)	Root	Diminished Triad (1–♭3–♭5)
C	C – Eb – Gb	F#/Gb	F# – A – C
C#/Db	Db – E – G	G	G – Bb – Db
D	D – F – Ab	G#/Ab	Ab – B – D
D#/Eb	Eb – Gb – A	A	A – C – Eb
E	E – G – Bb	A#/Bb	Bb – Db – E
F	F – Ab – B	B	B – D – F

Augmented Triads in All 12 Keys

Root	Augmented Triad (1–3–♯5)	Root	Augmented Triad (1–3–♯5)
C	C – E – G#	F#/Gb	F# – A# – D
C#/Db	Db – F – A	G	G – B – D#
D	D – F# – A#	G#/Ab	Ab – C – E
D#/Eb	Eb – G – B	A	A – C# – F
E	E – G# – C	A#/Bb	Bb – D – F#
F	F – A – C#	B	B – D# – G

Practice Tip

Notice how augmented triads "recycle" — because they are built from equally spaced Major 3rds, C augmented, E augmented, and G# augmented all share the exact same three notes. There are really only 4 unique augmented triads in total, unlike 12 unique Major, minor, and diminished triads.

Chapter Summary

- A Diminished triad follows 1–♭3–♭5 (3 + 3 semitones) — two stacked minor 3rds, creating a tense, symmetrical sound.
- An Augmented triad follows 1–3–♯5 (4 + 4 semitones) — two stacked Major 3rds, creating a floating, symmetrical sound.
- Diminished triads occur naturally in every major key as the vii° chord and typically resolve to I.
- Augmented triads are not naturally diatonic — they are introduced chromatically for tension, color, or as passing chords.
- Because of their symmetrical structure, augmented triads repeat every 4 semitones, meaning there are only 4 unique augmented triads total.

Practice Exercises

These exercises build both the hand-shapes and the ear-recognition for these two less-common but essential triad qualities.

Exercise 1: Building Diminished Triads by Semitone

- Starting on C, build a Diminished triad using semitone counting (3 semitones, then 3 more semitones).
- Repeat for D, F, and A.
- Cross-check each result against the reference table in section 5.3.

Self-check:

☐ I built all 4 diminished triads correctly using semitone counting.

☐ I can explain why a diminished triad sounds more "tense" than a minor triad.

Exercise 2: Building Augmented Triads by Semitone

- Starting on C, build an Augmented triad using semitone counting (4 semitones, then 4 more semitones).
- Repeat for Eb, F#, and A — then notice these four all share the exact same notes as C augmented.
- Cross-check each result against the reference table in section 5.3.

Self-check:

☐ I built the augmented triad correctly and confirmed the "repeats every 4 semitones" pattern myself.

☐ I can name the 4 unique augmented triads in total.

Exercise 3: Four Qualities, Same Root

- Pick a single root (e.g., D). Build and play, in order: D Major, D minor, D diminished, D augmented.
- Play them back to back, slowly, listening for exactly which note moves at each step.
- Repeat the same four-quality sequence starting on A and on F.

Self-check:

☐ I can build all 4 qualities from the same root without hesitating.

☐ I can identify exactly which note (3rd or 5th) changes between each pair.

Exercise 4: The vii° Resolution

- In the key of C Major, play the vii° chord (B diminished), then resolve it by moving to the I chord (C Major).
- Repeat this vii°→I resolution in G Major (F#dim → G) and in F Major (Edim → F).
- Notice how strongly the diminished chord seems to "want" to move to the I chord.

Self-check:

☐ I can play the vii°→I resolution correctly in at least 3 different keys.

☐ I can describe, in my own words, why the vii° chord feels unresolved on its own.

Exercise 5: Ear Training: Four-Way Identification

- Have someone else (or a recording/app) play triads in random order, drawn from all four qualities: Major, minor, diminished, augmented.
- For each one, identify which of the four qualities you heard.
- Review any that fooled you, and play those specific roots through all four qualities again slowly to re-anchor the sound.

Self-check:

☐ I correctly identified the quality at least 7 times out of 10 across all four types.

☐ I can describe, in my own words, the distinct "feeling" of each of the four triad qualities.

Next Chapter Preview

Chapter 6 will move beyond triads into 7th chords — adding a fourth note to each triad quality to build the richer, jazzier harmonic colors used throughout popular and jazz music.

Chapter 6 · Chord Inversions

So far, every chord in this book has been played in root position — root on the bottom, then 3rd, then 5th, stacked neatly upward. But a chord's identity does not depend on which note sits at the bottom; C Major is still C Major whether C, E, or G is the lowest-sounding note. This chapter introduces inversions: rearranging the same three notes to make chord progressions smoother, easier to play, and more musical.

6.1 Root Position (Review)

In root position, the root of the chord is the lowest note, with the 3rd and 5th stacked above it in order.

Position	Notes Bottom → Top (C Major)	Bass Note	Interval Pattern (bottom to top)
Root Position	C – E – G	C (the root)	3rd + 3rd (a "stacked thirds" shape)

> **Key Idea**
> Root position is the "textbook" shape you have used throughout Chapters 3–5. It is the clearest-sounding voicing and the correct starting point for learning any new chord — but it is rarely the most convenient shape for playing smooth progressions.

6.2 First Inversion

To build the first inversion, take the root position chord and move the root note up one octave — or equivalently, simply reorder the notes so the 3rd is now on the bottom.

The result, C Major first inversion, is spelled E – G – C, with the 3rd (E) as the lowest note.

Position	Notes Bottom → Top (C Major)	Bass Note	Interval Pattern (bottom to top)
1st Inversion	E – G – C	E (the 3rd)	3rd + 4th

> **How to Recognize It**
> A triad is in first inversion whenever its 3rd is the lowest note. A quick trick: look for the interval of a 4th between the top two notes (G to C above is a 4th) — root position and 2nd inversion do not have this shape.

6.3 Second Inversion

To build the second inversion, take the first inversion and move its lowest note (the original root's 3rd) up an octave again — or simply reorder the notes so the 5th is now on the bottom.

The result, C Major second inversion, is spelled G – C – E, with the 5th (G) as the lowest note.

Position	Notes Bottom → Top (C Major)	Bass Note	Interval Pattern (bottom to top)
2nd Inversion	G – C – E	G (the 5th)	4th + 3rd

All Three Positions Side by Side (C Major)

Position	Notes Bottom → Top	Bass Note	Quick Recognition
Root Position	C – E – G	C (root)	A "clean" stack of two 3rds
1st Inversion	E – G – C	E (3rd)	A 4th appears at the top
2nd Inversion	G – C – E	G (5th)	A 4th appears at the bottom

Common Confusion

The chord's NAME and ROOT never change across inversions — it is still "C Major" in all three cases. Only the BASS NOTE (the lowest-sounding note) changes. Musicians sometimes write this explicitly as a slash chord: C (root position), C/E (1st inversion), C/G (2nd inversion).

6.4 Smooth Movement Between Chords

The real reason inversions matter is voice leading — moving from one chord to the next using the smallest possible hand movement, so that progressions sound connected and flowing rather than jumpy.

The Problem With Root Position Only

If you play C Major, then F Major, then G Major, all in root position, your hand must leap across the keyboard for every single chord change — even though these chords share notes in common.

Progression Step	Root-Position-Only Notes	Hand Movement
C Major (root position)	C – E – G	—
F Major (root position)	F – A – C	Large leap upward
G Major (root position)	G – B – D	Large leap upward again

The Solution: Choose the Closest Inversion

Instead, choose the inversion of each chord that keeps the notes as close as possible to the previous chord — ideally, keep any shared or nearby notes in nearly the same hand position.

Progression Step	Smooth-Voice-Leading Notes	Hand Movement
C Major (root position)	C – E – G	—
F Major (2nd inversion)	C – F – A	Only 2 notes move a small step; C stays
G Major (2nd inversion)	D – G – B	Small stepwise motion from the previous chord

Key Idea

The goal of voice leading is simple: find the inversion of the next chord that shares notes with, or sits closest to, the current chord's notes. This is what makes a chord progression feel connected rather than like separate, disconnected shapes.

6.5 Inversions in All Keys

The same three-position pattern (root, 1st inversion, 2nd inversion) applies identically in every key — only the specific notes change. Use this table as a reference while practicing, the same way you have with previous chapters' key tables.

Key	Root Position	1st Inversion	2nd Inversion
C Major	C – E – G	E – G – C	G – C – E
G Major	G – B – D	B – D – G	D – G – B
D Major	D – F# – A	F# – A – D	A – D – F#
F Major	F – A – C	A – C – F	C – F – A
A Minor	A – C – E	C – E – A	E – A – C
E Minor	E – G – B	G – B – E	B – E – G

Practice Sequencing

Practice inversions the same way you practiced Major and minor triads in earlier chapters: work through all 12 Major keys (and, once comfortable, minor keys too) in circle-of-fifths order, playing root position → 1st inversion → 2nd inversion → back to root position, one octave up.

Chapter Summary

- Root position has the root as the lowest note; the chord's notes stack as two 3rds.
- First inversion moves the 3rd to the bottom; a 4th appears between the top two notes.
- Second inversion moves the 5th to the bottom; a 4th appears between the bottom two notes.
- A chord's name and root never change across inversions — only the bass note changes.
- Voice leading means choosing the inversion of each chord that keeps the hand as close as possible to the previous chord, producing smoother, more connected progressions.
- The same three-position pattern applies in every major and minor key.

Practice Exercises

Work through these in order — the final two exercises are where inversions really start to feel useful, so do not skip the earlier ones that build the raw hand-shape fluency first.

Exercise 1: Building All Three Positions

- Starting from C Major root position, build 1st inversion and 2nd inversion by hand, without looking at the reference table.
- Repeat the same process for F Major and G Major.
- For each chord, say out loud which note is now the bass note.

Self-check:

☐ I can build all 3 positions of C, F, and G Major without the reference table.

☐ I can correctly identify the bass note of each position on sight.

Exercise 2: Position Recognition Drill

- Without playing, look at a written chord (or have someone play one) and identify whether it is root position, 1st inversion, or 2nd inversion, using the "4th on top / 4th on bottom" trick from section 6.3.
- Check your answer, then play the chord to confirm.
- Repeat with at least 6 different chords across different keys.

Self-check:

☐ I correctly identified the inversion of at least 5 out of 6 chords.

☐ I can explain the "4th" trick in my own words.

Exercise 3: C–F–G Voice Leading

- Play C Major (root position) → F Major (root position) → G Major (root position) and notice the large hand leaps.
- Now play the same progression using the smooth-voice-leading inversions from section 6.4: C (root) → F (2nd inversion) → G (2nd inversion).
- Compare how much smaller the hand movement is in the second version.

Self-check:

☐ I can play both versions of the C-F-G progression.

☐ I can describe, in my own words, why the second version feels smoother.

Exercise 4: Build Your Own Smooth Progression

- Choose a simple 4-chord progression in a key of your choice (e.g., I–V–vi–IV from Chapter 2).
- For each chord, choose the inversion that keeps the hand closest to the previous chord's position — use trial and error at the keyboard.
- Play the full progression through several times until the hand movement feels minimal and natural.

Self-check:

☐ I built a 4-chord progression using inversions I chose myself, not copied from a table.

☐ My progression has noticeably smoother hand movement than an all-root-position version.

Exercise 5: All-Keys Inversion Fluency

- Pick a random Major key. As quickly as you can, play its root position, 1st inversion, and 2nd inversion triads in a row.
- Repeat for at least 6 different keys, gradually increasing speed.
- Finally, repeat the same drill using minor triads in at least 3 keys.

Self-check:

☐ I can play all 3 positions of a random Major key within 15 seconds.

☐ I can also do this for at least 3 minor keys.

Next Chapter Preview

Chapter 7 will build directly on inversions by introducing common chord progressions — how chords like I, IV, V, and vi combine into the patterns you hear across countless real songs.

Chapter 7 · Diatonic Chords

Chapter 2 introduced the idea that every major scale produces seven naturally occurring chords, labeled with Roman numerals. This chapter goes deeper: it groups those seven chords into functional "families," turns scale-to-chord conversion into a fast, repeatable skill, and — most importantly — shows how these chords combine into the common progressions you will recognize from real songs.

7.1 The Seven Diatonic Chords: I – ii – iii – IV – V – vi – vii°

Every major key produces exactly this sequence of chord qualities, built one on each scale degree:

Roman Numeral	Scale Degree	Quality	Key of C	Key of G	Key of D
I	1	Major	C	G	D
ii	2	minor	Dm	Am	Em
iii	3	minor	Em	Bm	F#m
IV	4	Major	F	C	G
V	5	Major	G	D	A
vi	6	minor	Am	Em	Bm
vii°	7	Diminished	Bdim	F#dim	C#dim

> **Quick Refresher**
> This Major–minor–minor–Major–Major–minor–diminished pattern was introduced in Chapter 2 — if it does not yet feel automatic, review section 2.4 before continuing, since the rest of this chapter builds directly on it.

7.2 Major-Key Chord Families

Beyond quality, each diatonic chord also has a harmonic function — a "job" it does within the key. Music theory groups the seven chords into three families based on shared notes and shared behavior.

Family	Chords	Function	Typical Behavior
Tonic	I, vi, (iii)	Home / rest	Feels stable and resolved; progressions usually start and end here
Subdominant	IV, ii	Movement away from home	Creates gentle motion, often leads toward the dominant
Dominant	V, vii°	Tension / pull back home	Creates strong pull that resolves naturally back to I

Chords within the same family can often substitute for one another because they share two of their three notes. For example, I (C–E–G) and vi (A–C–E) share the notes C and E, which is why vi is often called the "relative minor" and can stand in for I in a progression.

> **Key Idea**
> Thinking in families — Tonic, Subdominant, Dominant — rather than seven separate, unrelated chords is what lets experienced players understand WHY a progression works, and improvise reasonable substitutions on the spot.

7.3 Finding Chords From a Scale

This section turns "harmonizing a scale" (Chapter 2) into a fast mental shortcut you can apply on sight, in any key, without writing anything down.

The Fast Method

1. Spell the Major scale of the key you are working in (7 letters).
2. Assign Roman numerals I through vii° to each degree in order.
3. Apply the fixed quality pattern: Major, minor, minor, Major, Major, minor, diminished.
4. Build each triad by taking that scale degree, skipping one, taking the next, skipping one, taking the next — exactly as in Chapters 1 and 2.

Worked Example: Key of A Major

Step	Result
1. Spell the scale	A – B – C# – D – E – F# – G#
2. Assign numerals	I(A) ii(B) iii(C#) IV(D) V(E) vi(F#) vii°(G#)
3. Apply quality pattern	Major, minor, minor, Major, Major, minor, diminished
4. Build each chord	A, Bm, C#m, D, E, F#m, G#dim

> **Practice Tip**
> The single most valuable habit in this section is speed: the goal is not just to be ABLE to find the diatonic chords of a key, but to do it in under 20 seconds using the fast method — this becomes essential once you start learning songs by ear.

7.4 Common Chord Progressions

Certain sequences of diatonic chords appear again and again across pop, film, devotional, and folk music because their family movement (Tonic → Subdominant/Dominant → Tonic) sounds satisfying to the ear. Below are the progressions most worth learning first.

Progression	Roman Numerals	In C Major	In G Major	Common In
The "Pop" Progression	I – V – vi – IV	C – G – Am – F	G – D – Em – C	Countless modern pop and worship songs
The "50s" Progression	I – vi – IV – V	C – Am – F – G	G – Em – C – D	Doo-wop, classic pop ballads
The Basic Cadence	I – IV – V – I	C – F – G – C	G – C – D – G	Folk, hymns, blues turnarounds
The Circle Progression	vi – IV – I – V	Am – F – C – G	Em – C – G – D	Anthemic pop, rock choruses
The Descending Fall	I – V – IV – V	C – G – F – G	G – D – C – D	Classic rock, gospel-tinged pop

Why These Work

Every progression above moves logically between the Tonic, Subdominant, and Dominant families from section 7.2 — none of them are random. Once you can hear "which family" a chord belongs to, you can predict where a progression is likely headed even before you hear the next chord.

Chapter Summary

- Every major key produces the same seven diatonic chords: I, ii, iii, IV, V, vi, vii°, following the fixed Major-minor-minor-Major-Major-minor-diminished pattern.
- The seven chords group into three functional families: Tonic (home), Subdominant (movement away), and Dominant (tension that pulls back home).
- Chords sharing two notes (e.g., I and vi) can often substitute for one another within the same family.
- The fast method — spell the scale, number the degrees, apply the quality pattern, stack thirds — turns scale-to-chord conversion into a skill usable in seconds.
- Common progressions (I-V-vi-IV, I-vi-IV-V, I-IV-V-I, vi-IV-I-V, I-V-IV-V) recur across countless songs because they move sensibly between the three chord families.

Practice Exercises

These exercises build speed at finding diatonic chords and comfort playing the progressions that use them — two skills that reinforce each other directly.

Exercise 1: Speed Harmonization

- Using the fast method from section 7.3, find all seven diatonic chords of D Major, then A Major, then Eb Major.
- Time yourself on each key — aim for under 20 seconds once you know the method well.
- Check your answers against the reference table from Chapter 2, section 2.6.

Self-check:

☐ I correctly found all 7 chords in all 3 keys.

☐ My time per key is approaching or under 20 seconds.

Exercise 2: Family Sorting

- Without looking at section 7.2, try to sort the seven Roman numerals (I, ii, iii, IV, V, vi, vii°) into the three families: Tonic, Subdominant, Dominant.
- Check your answer against the table in section 7.2.
- For any you got wrong, play the two chords involved and listen for the shared notes that explain the grouping.

Self-check:

☐ I correctly sorted at least 5 of the 7 chords into their families.

☐ I can explain, using shared notes, why I and vi belong to the same family.

Exercise 3: Playing the Five Core Progressions

- In the key of C, play through all five progressions listed in section 7.4, using root position triads.
- Repeat the same five progressions in G Major.
- Notice which progressions sound most familiar to you, and try to recall a real song that uses each one.

Self-check:

☐ I can play all 5 progressions cleanly in both C and G Major.

☐ I identified at least 2 real songs that use one of these progressions.

Exercise 4: Progression With Smooth Voice Leading

- Choose one progression from section 7.4 in a key of your choice.
- Using the voice-leading skills from Chapter 6, find the smoothest inversion path through the whole progression.
- Play the progression through several times, aiming for minimal hand movement between chords.

Self-check:

☐ I chose inversions that noticeably reduced hand movement compared to all-root-position.

☐ I can play the full progression smoothly from memory.

Exercise 5: Compose Your Own 4-Chord Progression

- Using only diatonic chords from a single major key, write your own original 4-chord progression (it does not need to match one from section 7.4).
- Label each chord with its Roman numeral and its family (Tonic/Subdominant/Dominant).
- Play it through, then try the same numeral sequence transposed into two other keys.

Self-check:

☐ I created an original progression using correctly labeled diatonic chords.

☐ I successfully transposed it into at least 2 other keys.

Next Chapter Preview

Chapter 8 will build on this chapter's progressions by introducing 7th chords — adding a fourth note to

diatonic triads for richer, more sophisticated harmonic color.

Chapter 8 · Seventh Chords

Chapter 1 previewed the idea of "stacking one more third" on top of a triad to create richer color. This chapter makes that idea concrete: five distinct 7th-chord qualities, each with its own formula, sound, and typical use. Mastering these five shapes is the single biggest step between playing simple triads and playing like a real jazz, gospel, or contemporary keyboardist.

8.1 From Triad to 7th Chord

A 7th chord is a triad with one more third stacked on top — a fourth note added a 3rd above the 5th. That fourth note is called the 7th, and depending on its exact distance from the root, it creates one of five different chord qualities.

Chord Quality	Formula	Built From	Symbol Example
Major 7	1 – 3 – 5 – 7	Major triad + Major 7th	Cmaj7
Dominant 7	1 – 3 – 5 – ♭7	Major triad + minor 7th	C7
Minor 7	1 – ♭3 – 5 – ♭7	minor triad + minor 7th	Cm7
Half-Diminished 7	1 – ♭3 – ♭5 – ♭7	Diminished triad + minor 7th	Cm7♭5 (or Cø7)
Diminished 7	1 – ♭3 – ♭5 – 𝄫7	Diminished triad + diminished 7th	Cdim7 (or C°7)

Key Idea

Every 7th chord quality is simply "a triad you already know" plus one specific type of 7th on top. Learning 7th chords is mostly about learning which of two 7th intervals (Major 7th = 11 semitones, or minor 7th = 10 semitones) to add to each triad — not learning five unrelated new shapes from scratch.

8.2 Major 7 (maj7)

A Major 7 chord adds a Major 7th (11 semitones from the root) on top of a Major triad. Formula: 1–3–5–7.

Root	Major Triad	Add Major 7th	Full Chord (Cmaj7 example)
C	C – E – G	B (11 semitones up)	C – E – G – B

Character: smooth, dreamy, lush — often described as "jazzy" in a soft, relaxed way. Common in ballads, jazz standards, R&B, and lo-fi/chill styles, frequently used on the I and IV chords of a major key.

8.3 Dominant 7 (7)

A Dominant 7 chord adds a minor 7th (10 semitones from the root) on top of a Major triad. Formula: 1–3–5–♭7.

Root	Major Triad	Add minor 7th	Full Chord (C7 example)
C	C – E – G	Bb (10 semitones up)	C – E – G – Bb

Character: bluesy, gritty, unresolved — strongly wants to move somewhere else. This is the chord built naturally on the V degree of a major scale (see Chapter 7), which is why it is called "dominant": it creates the pull that resolves back to the I chord. Essential in blues, gospel, funk, and jazz.

> **Common Confusion**
> "Dominant 7" does NOT mean "Major 7." A plain "7" symbol (like C7 or G7) always means Dominant 7 (minor 7th on top); "maj7" must be written out explicitly for a Major 7 chord. Mixing these two up is one of the most common beginner errors in chord reading.

8.4 Minor 7 (m7)

A Minor 7 chord adds a minor 7th (10 semitones from the root) on top of a minor triad. Formula: 1–♭3–5–♭7.

Root	minor Triad	Add minor 7th	Full Chord (Cm7 example)
C	C – Eb – G	Bb (10 semitones up)	C – Eb – G – Bb

Character: warm, mellow, and softer than a plain minor triad — the added 7th smooths out the darker minor color. Extremely common in jazz, neo-soul, and pop, especially on the ii and vi chords of a progression (e.g., ii7–V7–Imaj7).

8.5 Half-Diminished 7 (m7♭5 / ø7)

A Half-Diminished 7 chord adds a minor 7th (10 semitones from the root) on top of a Diminished triad. Formula: 1–♭3–♭5–♭7.

Root	Diminished Triad	Add minor 7th	Full Chord (Cm7♭5 example)
C	C – Eb – Gb	Bb (10 semitones up)	C – Eb – Gb – Bb

Character: tense but slightly softer than a fully diminished chord — "half" diminished because only the triad portion is diminished, while the 7th is a minor (not doubly-lowered) 7th. This is the chord built naturally on the vii° degree of a major scale when extended to four notes, and is a key building block of minor-key ii–V–i progressions in jazz.

8.6 Diminished 7 (dim7 / °7)

A fully Diminished 7 chord adds a diminished 7th (9 semitones from the root — sometimes written ♭♭7) on top of a Diminished triad. Formula: 1–♭3–♭5–♭♭7.

Root	Diminished Triad	Add diminished 7th	Full Chord (Cdim7 example)
C	C – Eb – Gb	Bbb / A (9 semitones up)	C – Eb – Gb – A

Character: highly tense, dramatic, and — unlike the other 7th chords — perfectly symmetrical (built from four stacked minor 3rds). This symmetry means a diminished 7th chord repeats every 3 semitones, so there are only

3 unique diminished 7th chords in total. Common in classical, film-score, and jazz writing as a dramatic passing or transitional chord.

> **Practice Tip**
> Because it is built from stacked minor 3rds (3+3+3+3 semitones), a diminished 7th chord's notes can each act as the "root" of the identical chord shape — C°7, Eb°7, Gb°7, and A°7 are all the exact same four notes.

All Five 7th-Chord Qualities Side by Side (Root: C)

Quality	Formula	Notes	Character
Major 7	1-3-5-7	C-E-G-B	Smooth, dreamy
Dominant 7	1-3-5-♭7	C-E-G-Bb	Bluesy, unresolved
Minor 7	1-♭3-5-♭7	C-Eb-G-Bb	Warm, mellow
Half-Dim 7	1-♭3-♭5-♭7	C-Eb-Gb-Bb	Tense, jazzy
Diminished 7	1-♭3-♭5-♭♭7	C-Eb-Gb-A	Highly tense, symmetrical

Chapter Summary

- A 7th chord is a triad plus one more stacked 3rd, adding a fourth note called the 7th.
- Major 7 (1-3-5-7): Major triad + Major 7th — smooth, dreamy sound.
- Dominant 7 (1-3-5-♭7): Major triad + minor 7th — bluesy, built on the V chord, wants to resolve.
- Minor 7 (1-♭3-5-♭7): minor triad + minor 7th — warm, mellow, common on ii and vi chords.
- Half-Diminished 7 (1-♭3-♭5-♭7): Diminished triad + minor 7th — tense, common on vii° in minor-key ii-V-i progressions.
- Diminished 7 (1-♭3-♭5-♭♭7): Diminished triad + diminished 7th — fully symmetrical, only 3 unique voicings exist.

Practice Exercises

Work through each quality individually before mixing them — these five shapes are easy to confuse at first, so accuracy matters more than speed early on.

Exercise 1: Building Each Quality on C

- Starting from a C Major triad, add each of the five 7th-chord qualities in turn: Cmaj7, C7, Cm7, Cm7♭5, Cdim7.
- For each one, say the full formula out loud as you play it (e.g., "1, 3, 5, flat 7" for C7).
- Play all five back to back, slowly, and listen for how each one is subtly different.

Self-check:

☐ I built all 5 seventh-chord qualities correctly on C.

☐ I can state each chord's formula from memory.

Exercise 2: Dominant 7 vs. Major 7 Discrimination

- Play Cmaj7, then C7, back and forth several times, listening specifically to the 7th note (B vs Bb).
- Repeat the same comparison on G (Gmaj7 vs G7) and F (Fmaj7 vs F7).
- Without looking, have someone play one or the other and try to identify which quality it is.

Self-check:

☐ I can reliably tell maj7 from dominant 7 by ear.

☐ I no longer confuse the "7" symbol with "maj7" when reading chord charts.

Exercise 3: ii7–V7–Imaj7 in Three Keys

- In C Major, play the progression Dm7 – G7 – Cmaj7 (the classic jazz "two-five-one").
- Repeat the same ii7-V7-Imaj7 progression in F Major (Gm7-C7-Fmaj7) and in G Major (Am7-D7-Gmaj7).
- Try adding smooth voice leading (Chapter 6) between the three chords.

Self-check:

☐ I can play the ii7-V7-Imaj7 progression in all 3 keys.

☐ I used at least one inversion to smooth out the hand movement.

Exercise 4: Half-Diminished and Diminished 7 Focus

- Build Bm7♭5 and Bdim7 (both based on the vii° of C Major) and compare them directly.
- Notice which single note differs between the two chords.
- Play Bm7♭5 resolving to Cmaj7, then Bdim7 resolving to C Major, and compare the two resolutions.

Self-check:

☐ I can build both Bm7♭5 and Bdim7 correctly and explain the one-note difference.

☐ I can describe how each one resolves differently to C.

Exercise 5: Ear Training: Five-Way Identification

- Have someone else (or a recording/app) play 10 seventh chords in random order, drawn from all five qualities covered in this chapter.
- For each one, try to identify the quality: Major 7, Dominant 7, Minor 7, Half-Diminished 7, or Diminished 7.
- Review any that fooled you, and play those specific chords again slowly, comparing them to their closest "neighbor" quality.

Self-check:

☐ I correctly identified the quality at least 6 times out of 10 across all five types.

☐ I can describe, in my own words, the distinct character of each of the five seventh-chord qualities.

Next Chapter Preview

Chapter 9 will move beyond 7th chords into extended chords — 9ths, 11ths, and 13ths — for the full, rich harmonic palette used in jazz and contemporary keyboard playing.

Chapter 9 · Common Chord Progressions

Chapter 7 introduced several progressions as an application of diatonic chord families. This chapter goes further: it treats five essential progressions — including the famous 12-bar blues — as complete, independently practiced pieces of vocabulary, and drills each one across multiple keys so you can play it instantly whenever a song calls for it.

9.1 I – IV – V: The Foundational Progression

I–IV–V is the single most fundamental progression in Western music — nearly every folk song, hymn, and simple rock song leans on it in some form. It moves from home (I), briefly away (IV), through tension (V), and (usually) back home again.

Key	I	IV	V
C	C	F	G
G	G	C	D
D	D	G	A
F	F	Bb	C
A	A	D	E

> **Why It Works**
> I, IV, and V are the three Major chords in every major key (see Chapter 7) — none of them require any accidentals, and each one is a Perfect 4th or 5th apart from the others, so the hand movement is naturally comfortable.

9.2 I – V – vi – IV: The "Pop" Progression

This four-chord loop is arguably the most recognizable chord progression in modern popular music, appearing across pop, worship, rock, and film songs from many different decades.

Key	I	V	vi	IV
C	C	G	Am	F
G	G	D	Em	C
D	D	A	Bm	G
A	A	E	F#m	D
E	E	B	C#m	A

Notice this progression loops naturally — after IV, it can return smoothly to I to repeat, which is exactly why it works so well for entire song sections rather than just a single cadence.

9.3 ii – V – I: The Jazz Standard

ii–V–I (often written "two-five-one") is the backbone of jazz harmony. Where I–IV–V uses only Major chords, ii–V–I introduces a minor chord (ii) leading into the dominant (V) before resolving to home (I) — frequently using 7th chords for a smoother, jazzier sound.

Key	ii	V	I	With 7ths
C	Dm	G	C	Dm7 – G7 – Cmaj7
F	Gm	C	F	Gm7 – C7 – Fmaj7
Bb	Cm	F	Bb	Cm7 – F7 – Bbmaj7
G	Am	D	G	Am7 – D7 – Gmaj7

> **Key Idea**
> ii–V–I appears so frequently in jazz standards that many musicians practice it in all 12 keys as a dedicated daily exercise — much like scales. Learning to move through it fluently is one of the fastest ways to sound more "jazz-fluent" at the keyboard.

9.4 I – vi – IV – V: The "50s" Progression

This progression predates and heavily influenced I-V-vi-IV; it was the backbone of doo-wop and early rock 'n' roll, and still appears constantly in ballads and retro-styled songs today.

Key	I	vi	IV	V
C	C	Am	F	G
G	G	Em	C	D
F	F	Dm	Bb	C
D	D	Bm	G	A

Compare this directly to section 9.2's I–V–vi–IV: the same four chords appear, simply reordered. This is a good demonstration of how a small set of diatonic chords can be rearranged into multiple distinct, recognizable progressions.

9.5 The 12-Bar Blues

The 12-bar blues is a complete, self-contained harmonic form — 12 measures long, built almost entirely from the I, IV, and V chords (usually as dominant 7th chords), that repeats as a full cycle throughout a blues, rock, or boogie-woogie song.

The Standard 12-Bar Blues Pattern (Key of C)

Bar	1	2	3	4	5	6	7	8	9	10	11	12
Chord	C7	C7	C7	C7	F7	F7	C7	C7	G7	F7	C7	G7

In Roman numerals, this is: I7 I7 I7 I7 / IV7 IV7 I7 I7 / V7 IV7 I7 V7 — four bars of I7, two bars of IV7, two bars back to I7, then a final "turnaround" of V7-IV7-I7-V7 that leads the progression back to the start.

Key	I7 (bars 1-4, 7-8, 11)	IV7 (bars 5-6, 10)	V7 (bars 9, 12)
C	C7	F7	G7
G	G7	C7	D7
A	A7	D7	E7
E	E7	A7	B7

Practice Tip

Play through the full 12 bars with a steady beat (even just counting "1-2-3-4" out loud per bar) rather than treating each chord in isolation — the 12-bar blues is meant to be felt as one continuous cyclical form, not a list of separate chords.

9.6 Progressions in Different Keys

Every progression in this chapter is written using Roman numerals precisely so it can be transposed instantly. The process is always the same: find the diatonic chords of your target key (Chapter 7, section 7.3), then substitute them in for the numerals.

Transposition Reminder

To move any progression into a new key: (1) spell that key's Major scale, (2) find its I, ii, IV, V, and vi chords using the fixed quality pattern from Chapter 7, (3) substitute those chords in for the Roman numerals, keeping the order identical.

Chapter Summary

- I–IV–V is the most foundational progression, using only the three Major chords of a key.
- I–V–vi–IV ("Pop") and I–vi–IV–V ("50s") use the same four chords in different orders, each with a distinct feel.
- ii–V–I is the backbone of jazz harmony, often played with 7th chords (ii7–V7–Imaj7) for a smoother sound.
- The 12-bar blues is a complete 12-measure cyclical form built from I7, IV7, and V7, following a fixed bar-by-bar pattern.
- Because every progression is described in Roman numerals, all of them transpose instantly into any key using the diatonic-chord method from Chapter 7.

Practice Exercises

These exercises move from single progressions in isolation toward switching between progressions fluently and applying the 12-bar blues as a complete form.

Exercise 1: The Big Four, One Key at a Time

- In the key of C, play I-IV-V, then I-V-vi-IV, then I-vi-IV-V, then ii-V-I, back to back.
- Say the Roman numerals out loud as you play each chord.
- Repeat the same four progressions in G Major.

Self-check:

☐ I can play all 4 progressions cleanly in both C and G Major.

☐ I can say the Roman numerals from memory while playing.

Exercise 2: Compare and Contrast: Pop vs. 50s

- Play I-V-vi-IV and I-vi-IV-V back to back in the same key, noticing that they use identical chords in a different order.
- Try to identify a song (in your own memory) that matches each ordering.
- Repeat the comparison in a second key of your choice.

Self-check:

☐ I can play both progressions and clearly hear the difference despite using the same 4 chords.

☐ I identified at least one real song matching each progression.

Exercise 3: ii–V–I With 7th Chords, Three Keys

- Play Dm7-G7-Cmaj7 in C Major, using inversions from Chapter 6 for smooth voice leading.
- Repeat in F Major (Gm7-C7-Fmaj7) and Bb Major (Cm7-F7-Bbmaj7).
- Try playing the same ii-V-I twice in a row, moving down a whole step for the second repetition, as jazz musicians commonly practice it.

Self-check:

☐ I can play ii7-V7-Imaj7 smoothly in at least 3 keys.

☐ I successfully moved the progression down a whole step and repeated it.

Exercise 4: 12-Bar Blues in C

- Play through the full 12-bar blues pattern in C, using the bar-by-bar chart in section 9.5, keeping a steady beat.
- Repeat the full 12 bars at least 3 times in a row without stopping, as if looping through a song.
- Try adding simple left-hand root notes on beat 1 of each bar while your right hand plays the chords.

Self-check:

☐ I can play the complete 12-bar blues in C without stopping.

☐ I successfully looped it at least 3 times in a row.

Exercise 5: 12-Bar Blues Transposition Challenge

- Using the transposition method from section 9.6, work out the 12-bar blues in G Major on paper first.
- Play through your G Major 12-bar blues and check it against the table in section 9.5.
- Finally, choose one more key of your own choice and repeat the full process independently.

Self-check:

☐ I correctly transposed the 12-bar blues into G Major by hand before checking the table.

☐ I completed a third key entirely on my own.

Next Chapter Preview

Chapter 10 will shift focus from WHAT chords to play to HOW to play them well — metronome practice, smooth transitions, rhythm patterns, and hand coordination.

Chapter 10 · Chord Progression Practice

Every chapter so far has focused on WHAT to play — which notes make up a chord, and which chords make up a progression. This chapter focuses on HOW to play those progressions musically: with steady timing, smooth transitions, rhythmic interest, and coordinated hands. These are the skills that separate a player who "knows the chords" from a player who can actually perform them.

10.1 Playing With a Metronome

A metronome produces a steady, even click at a chosen tempo (measured in beats per minute, or BPM). Practicing with one is the single most effective habit for developing reliable timing — internal timing that feels natural even without the click.

How to Set Up Metronome Practice

- Start slow: choose a tempo where you can play every chord change accurately, even if it feels too easy. 60–70 BPM is a good starting point for new progressions.
- Lock to the click: your chord changes (or your first beat of each new chord) should land exactly on a metronome click, not slightly before or after.
- Increase gradually: once a tempo feels completely comfortable and accurate, raise it by 5–10 BPM. Never jump straight to a fast tempo.
- Use subdivision clicks: many metronomes can click on every beat, or subdivide into eighth-notes — using subdivisions helps you feel the space between chord changes more precisely.

> **Key Idea**
> The goal of metronome practice is not to play "with" the click forever — it is to internalize a steady pulse so reliably that your timing stays even even when the metronome is turned off.

10.2 Changing Chords Smoothly

Smooth chord changes combine two things you have already studied — inversions and voice leading (Chapter 6) — with physical technique for the moment of transition itself.

Technique for Smooth Transitions

1. Look ahead: before you finish playing the current chord, your eyes (and mental attention) should already be identifying the next chord's shape.
2. Release and land together: lift all fingers of the current chord at the same moment, and land on the new chord's notes at the same moment — avoid a "rolled" or staggered change unless it is intentional.
3. Minimize finger travel: choose the inversion (Chapter 6) that keeps fingers as close as possible to their next position.
4. Keep a relaxed wrist: tension in the wrist or forearm is the most common cause of clumsy, delayed chord changes.

Common Mistake

Rushing the chord change to "catch up" after a late start is a common beginner habit that actually makes timing worse. If you are behind, it is almost always better to accept a slightly late chord and stay steady from that point forward, rather than to rush and destabilize the whole pulse.

10.3 Rhythm Patterns

A chord progression can be played in many different rhythmic styles using the exact same chords. Below are four foundational patterns worth practicing on every progression you learn.

Pattern Name	Description	Typical Feel
Block Chords	All notes of the chord played together, held for the full duration	Simple, hymn-like, foundational
Quarter-Note Pulse	Chord replayed (re-struck) on every beat	Driving, steady, rock/pop
Broken Chord (Arpeggio)	Notes of the chord played one at a time in sequence, rather than together	Flowing, ballad-like, gentle
Syncopated Comping	Chord struck on off-beats or unexpected beats within the bar	Jazzy, rhythmic, groove-based

Example (block chords vs. quarter-note pulse), one bar of C Major in 4/4 time: Block chords = play C-E-G once and hold for all 4 beats. Quarter-note pulse = re-strike C-E-G on beats 1, 2, 3, and 4.

Practice Tip

Take any single progression you already know (like I-V-vi-IV from Chapter 9) and play it through using all four rhythm patterns in a row. This reveals how much musical variety is possible from a fixed set of chords.

10.4 Left-Hand / Right-Hand Coordination

A common, versatile approach on keyboard is to split roles between the hands: the left hand outlines the harmony's foundation (often just the root, or root and 5th), while the right hand plays the fuller chord voicing or a rhythmic pattern.

A Simple Coordination Framework

Left Hand	Right Hand	Result
Root note only, on beat 1	Full chord, block or arpeggiated	Clean, simple, beginner-friendly texture
Root + 5th, alternating	Full chord, quarter-note pulse	Fuller, "walking bass" feel
Root, quarter notes	Chord on off-beats (syncopated)	Groove-oriented, pop/R&B feel

Start by practicing each hand completely separately until it is automatic, then combine them slowly, adding the metronome only once both hands feel confident independently.

Common Mistake

Trying to combine both hands at full speed before either hand is independently solid almost always leads to one hand "dragging" the other. Always separate, then slow-combine, then speed up — never skip straight to fast, both-hands-together practice.

10.5 Progression Exercises

The exercises below apply metronome discipline, smooth transitions, rhythm variety, and hand coordination directly to the progressions you already know from Chapters 7 and 9.

Chapter Summary

- Metronome practice builds reliable internal timing — start slow, lock chord changes to the click, and increase tempo gradually.
- Smooth chord changes combine looking ahead, releasing and landing together, minimizing finger travel via inversions, and staying relaxed.
- The same progression can be played with different rhythm patterns — block chords, quarter-note pulse, broken chords, and syncopated comping — for very different musical feels.
- Splitting roles between hands (left hand = foundation, right hand = harmony/rhythm) is a versatile, widely used keyboard texture.
- Always build coordination in stages: hands separately, then slow together, then gradually increase speed.

Practice Exercises

Use a real metronome (a phone app is fine) for every exercise in this set — timing is the whole point of this chapter.

Exercise 1: Metronome Foundation

- Set a metronome to 60 BPM. Play the I-IV-V progression (Chapter 9) in C Major, one chord per 4 beats, in block chords.
- Make sure each chord change lands exactly on beat 1, not before or after.
- Once perfectly steady, increase to 70 BPM, then 80 BPM.

Self-check:

☐ I can play I-IV-V at 80 BPM with every chord change landing exactly on the beat.

☐ I did not rush or drag at any point during the exercise.

Exercise 2: Smooth Transition Drill

- Using the I-V-vi-IV progression, choose smooth-voice-leading inversions (Chapter 6) for each chord.
- Practice just the transitions — play each chord for only 1 beat, then move to the next, focusing entirely on a clean, together release-and-land motion.
- Gradually increase tempo while keeping every transition clean.

Self-check:

☐ My chord changes are clean and simultaneous, without staggered or rolled notes.

☐ I successfully increased tempo without transitions becoming sloppy.

Exercise 3: Four Rhythms, One Progression

- Take the ii-V-I progression (Chapter 9) in F Major and play it through 4 times, once in each rhythm pattern from section 10.3: block chords, quarter-note pulse, broken chord, syncopated comping.
- Use the same tempo for all four so you can compare them directly.
- Choose your favorite of the four and explain why, in your own words.

Self-check:

☐ I performed the same progression in all 4 rhythm patterns.

☐ I can describe how the feel changed between patterns.

Exercise 4: Hands-Separate-Then-Together Coordination

- Using the 12-bar blues in C (Chapter 9), practice the left-hand root-note pattern alone until it is completely steady with the metronome.
- Separately, practice the right-hand chord pattern (using dominant 7th chords) alone until it is steady.
- Combine both hands at a slow tempo, only increasing speed once the combination feels solid.

Self-check:

☐ I practiced each hand separately before combining them.

☐ I can play the full 12-bar blues with both hands together at a steady tempo.

Exercise 5: Full Progression Performance

- Choose any progression from Chapter 9 and a rhythm pattern from section 10.3 of your choice.
- Set a metronome at a comfortable tempo and perform the progression from start to finish, at least 4 times through, without stopping.
- Record yourself if possible, and listen back specifically for timing consistency and clean chord changes.

Self-check:

☐ I performed a full progression 4 times through without stopping.

☐ On playback (or self-assessment), my timing stayed consistent throughout.

Next Chapter Preview

Chapter 11 turns everything learned so far into a per-key practice reference — for each key, working through Scale → Chord Family → Triads → Inversions → Progressions in one place, starting with C Major.

Chapter 11 · Chords by Key

This chapter is a hands-on practice reference, organized key by key. For each key, you will work through the same five-step sequence — Scale → Chord Family → Triads → Inversions → Progressions — pulling together everything learned in Chapters 1 through 10 into one focused practice routine per key. More keys will be added to this chapter over time; begin with C Major below.

How to Use This Chapter

For each key, work through the five sections in order: play the scale first to warm up and confirm the key's notes, then build the chord family, then drill the individual triads, then their inversions, and finally the progressions — which combine everything above into real musical practice.

11.1 Key of C Major

Scale

The C Major scale: C – D – E – F – G – A – B – C.

Chord Family (Diatonic Chords)

Numeral	Chord	Notes	Quality
I	C	C – E – G	Major
ii	Dm	D – F – A	minor
iii	Em	E – G – B	minor
IV	F	F – A – C	Major
V	G	G – B – D	Major
vi	Am	A – C – E	minor
vii°	Bdim	B – D – F	Diminished

Triads to Drill (Root Position)

Chord	Notes (Root Position)
C (I)	C – E – G
Dm (ii)	D – F – A
Em (iii)	E – G – B
F (IV)	F – A – C
G (V)	G – B – D
Am (vi)	A – C – E
Bdim (vii°)	B – D – F

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
C (I)	C–E–G	E–G–C	G–C–E
F (IV)	F–A–C	A–C–F	C–F–A
G (V)	G–B–D	B–D–G	D–G–B
Am (vi)	A–C–E	C–E–A	E–A–C

Progressions in This Key

Progression	Roman Numerals	Chords
I – IV – V	I – IV – V	C – F – G
Pop Progression	I – V – vi – IV	C – G – Am – F
50s Progression	I – vi – IV – V	C – Am – F – G
Jazz ii–V–I	ii7 – V7 – Imaj7	Dm7 – G7 – Cmaj7
12-Bar Blues (I7-IV7-V7)	I7 – IV7 – V7	C7 – F7 – G7

11.2 Key of G Major

Scale

The G Major scale: G – A – B – C – D – E – F# – G.

Chord Family (Diatonic Chords)

Numeral	Chord	Notes	Quality
I	G	G – B – D	Major
ii	Am	A – C – E	minor
iii	Bm	B – D – F#	minor
IV	C	C – E – G	Major
V	D	D – F# – A	Major
vi	Em	E – G – B	minor
vii°	F#dim	F# – A – C	Diminished

Triads to Drill (Root Position)

Chord	Notes (Root Position)
G (I)	G – B – D
Am (ii)	A – C – E

Chord	Notes (Root Position)
Bm (iii)	B – D – F#
C (IV)	C – E – G
D (V)	D – F# – A
Em (vi)	E – G – B
F#dim (vii°)	F# – A – C

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
G (I)	G–B–D	B–D–G	D–G–B
C (IV)	C–E–G	E–G–C	G–C–E
D (V)	D–F#–A	F#–A–D	A–D–F#
Em (vi)	E–G–B	G–B–E	B–E–G

Progressions in This Key

Progression	Roman Numerals	Chords
I – IV – V	I – IV – V	G – C – D
Pop Progression	I – V – vi – IV	G – D – Em – C
50s Progression	I – vi – IV – V	G – Em – C – D
Jazz ii–V–I	ii7 – V7 – Imaj7	Am7 – D7 – Gmaj7
12-Bar Blues (I7-IV7-V7)	I7 – IV7 – V7	G7 – C7 – D7

11.3 Key of D Major

Scale

The D Major scale: D – E – F# – G – A – B – C# – D.

Chord Family (Diatonic Chords)

Numeral	Chord	Notes	Quality
I	D	D – F# – A	Major
ii	Em	E – G – B	minor
iii	F#m	F# – A – C#	minor
IV	G	G – B – D	Major
V	A	A – C# – E	Major

Numeral	Chord	Notes	Quality
vi	Bm	B – D – F#	minor
vii°	C#dim	C# – E – G	Diminished

Triads to Drill (Root Position)

Chord	Notes (Root Position)
D (I)	D – F# – A
Em (ii)	E – G – B
F#m (iii)	F# – A – C#
G (IV)	G – B – D
A (V)	A – C# – E
Bm (vi)	B – D – F#
C#dim (vii°)	C# – E – G

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
D (I)	D–F#–A	F#–A–D	A–D–F#
G (IV)	G–B–D	B–D–G	D–G–B
A (V)	A–C#–E	C#–E–A	E–A–C#
Bm (vi)	B–D–F#	D–F#–B	F#–B–D

Progressions in This Key

Progression	Roman Numerals	Chords
I – IV – V	I – IV – V	D – G – A
Pop Progression	I – V – vi – IV	D – A – Bm – G
50s Progression	I – vi – IV – V	D – Bm – G – A
Jazz ii–V–I	ii7 – V7 – Imaj7	Em7 – A7 – Dmaj7
12-Bar Blues (I7-IV7-V7)	I7 – IV7 – V7	D7 – G7 – A7

11.4 Key of A Major

Scale

The A Major scale: A – B – C# – D – E – F# – G# – A.

Chord Family (Diatonic Chords)

Numeral	Chord	Notes	Quality
I	A	A – C# – E	Major
ii	Bm	B – D – F#	minor
iii	C#m	C# – E – G#	minor
IV	D	D – F# – A	Major
V	E	E – G# – B	Major
vi	F#m	F# – A – C#	minor
vii°	G#dim	G# – B – D	Diminished

Triads to Drill (Root Position)

Chord	Notes (Root Position)
A (I)	A – C# – E
Bm (ii)	B – D – F#
C#m (iii)	C# – E – G#
D (IV)	D – F# – A
E (V)	E – G# – B
F#m (vi)	F# – A – C#
G#dim (vii°)	G# – B – D

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
A (I)	A–C#–E	C#–E–A	E–A–C#
D (IV)	D–F#–A	F#–A–D	A–D–F#
E (V)	E–G#–B	G#–B–E	B–E–G#
F#m (vi)	F#–A–C#	A–C#–F#	C#–F#–A

Progressions in This Key

Progression	Roman Numerals	Chords
I – IV – V	I – IV – V	A – D – E
Pop Progression	I – V – vi – IV	A – E – F#m – D
50s Progression	I – vi – IV – V	A – F#m – D – E

Progression	Roman Numerals	Chords
Jazz ii–V–I	ii7 – V7 – Imaj7	Bm7 – E7 – Amaj7
12-Bar Blues (I7-IV7-V7)	I7 – IV7 – V7	A7 – D7 – E7

11.5 Key of E Major

Scale

The E Major scale: E – F# – G# – A – B – C# – D# – E.

Chord Family (Diatonic Chords)

Numeral	Chord	Notes	Quality
I	E	E – G# – B	Major
ii	F#m	F# – A – C#	minor
iii	G#m	G# – B – D#	minor
IV	A	A – C# – E	Major
V	B	B – D# – F#	Major
vi	C#m	C# – E – G#	minor
vii°	D#dim	D# – F# – A	Diminished

Triads to Drill (Root Position)

Chord	Notes (Root Position)
E (I)	E – G# – B
F#m (ii)	F# – A – C#
G#m (iii)	G# – B – D#
A (IV)	A – C# – E
B (V)	B – D# – F#
C#m (vi)	C# – E – G#
D#dim (vii°)	D# – F# – A

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
E (I)	E–G#–B	G#–B–E	B–E–G#
A (IV)	A–C#–E	C#–E–A	E–A–C#
B (V)	B–D#–F#	D#–F#–B	F#–B–D#

Chord	Root Position	1st Inversion	2nd Inversion
C#m (vi)	C#–E–G#	E–G#–C#	G#–C#–E

Progressions in This Key

Progression	Roman Numerals	Chords
I – IV – V	I – IV – V	E – A – B
Pop Progression	I – V – vi – IV	E – B – C#m – A
50s Progression	I – vi – IV – V	E – C#m – A – B
Jazz ii–V–I	ii7 – V7 – Imaj7	F#m7 – B7 – Emaj7
12-Bar Blues (I7-IV7-V7)	I7 – IV7 – V7	E7 – A7 – B7

11.6 Key of B Major

Scale

The B Major scale: B – C# – D# – E – F# – G# – A# – B.

Chord Family (Diatonic Chords)

Numeral	Chord	Notes	Quality
I	B	B – D# – F#	Major
ii	C#m	C# – E – G#	minor
iii	D#m	D# – F# – A#	minor
IV	E	E – G# – B	Major
V	F#	F# – A# – C#	Major
vi	G#m	G# – B – D#	minor
vii°	A#dim	A# – C# – E	Diminished

Triads to Drill (Root Position)

Chord	Notes (Root Position)
B (I)	B – D# – F#
C#m (ii)	C# – E – G#
D#m (iii)	D# – F# – A#
E (IV)	E – G# – B
F# (V)	F# – A# – C#
G#m (vi)	G# – B – D#

Chord	Notes (Root Position)
A#dim (vii°)	A# – C# – E

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
B (I)	B–D#–F#	D#–F#–B	F#–B–D#
E (IV)	E–G#–B	G#–B–E	B–E–G#
F# (V)	F#–A#–C#	A#–C#–F#	C#–F#–A#
G#m (vi)	G#–B–D#	B–D#–G#	D#–G#–B

Progressions in This Key

Progression	Roman Numerals	Chords
I – IV – V	I – IV – V	B – E – F#
Pop Progression	I – V – vi – IV	B – F# – G#m – E
50s Progression	I – vi – IV – V	B – G#m – E – F#
Jazz ii–V–I	ii7 – V7 – Imaj7	C#m7 – F#7 – Bmaj7
12-Bar Blues (I7-IV7-V7)	I7 – IV7 – V7	B7 – E7 – F#7

11.7 Key of F# Major

Scale

The F# Major scale: F# – G# – A# – B – C# – D# – E# – F#.

Chord Family (Diatonic Chords)

Numeral	Chord	Notes	Quality
I	F#	F# – A# – C#	Major
ii	G#m	G# – B – D#	minor
iii	A#m	A# – C# – E#	minor
IV	B	B – D# – F#	Major
V	C#	C# – E# – G#	Major
vi	D#m	D# – F# – A#	minor
vii°	E#dim	E# – G# – B	Diminished

Triads to Drill (Root Position)

Chord	Notes (Root Position)
F# (I)	F# – A# – C#
G#m (ii)	G# – B – D#
A#m (iii)	A# – C# – E#
B (IV)	B – D# – F#
C# (V)	C# – E# – G#
D#m (vi)	D# – F# – A#
E#dim (vii°)	E# – G# – B

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
F# (I)	F#–A#–C#	A#–C#–F#	C#–F#–A#
B (IV)	B–D#–F#	D#–F#–B	F#–B–D#
C# (V)	C#–E#–G#	E#–G#–C#	G#–C#–E#
D#m (vi)	D#–F#–A#	F#–A#–D#	A#–D#–F#

Progressions in This Key

Progression	Roman Numerals	Chords
I – IV – V	I – IV – V	F# – B – C#
Pop Progression	I – V – vi – IV	F# – C# – D#m – B
50s Progression	I – vi – IV – V	F# – D#m – B – C#
Jazz ii–V–I	ii7 – V7 – Imaj7	G#m7 – C#7 – F#maj7
12-Bar Blues (I7-IV7-V7)	I7 – IV7 – V7	F#7 – B7 – C#7

11.8 Key of D♭ Major

Scale

The D♭ Major scale: Db – Eb – F – Gb – Ab – Bb – C – Db.

Chord Family (Diatonic Chords)

Numeral	Chord	Notes	Quality
I	Db	Db – F – Ab	Major
ii	Ebm	Eb – Gb – Bb	minor

Numeral	Chord	Notes	Quality
iii	Fm	F – Ab – C	minor
IV	Gb	Gb – Bb – Db	Major
V	Ab	Ab – C – Eb	Major
vi	Bbm	Bb – Db – F	minor
vii°	Cdim	C – Eb – Gb	Diminished

Triads to Drill (Root Position)

Chord	Notes (Root Position)
Db (I)	Db – F – Ab
Ebm (ii)	Eb – Gb – Bb
Fm (iii)	F – Ab – C
Gb (IV)	Gb – Bb – Db
Ab (V)	Ab – C – Eb
Bbm (vi)	Bb – Db – F
Cdim (vii°)	C – Eb – Gb

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
Db (I)	Db–F–Ab	F–Ab–Db	Ab–Db–F
Gb (IV)	Gb–Bb–Db	Bb–Db–Gb	Db–Gb–Bb
Ab (V)	Ab–C–Eb	C–Eb–Ab	Eb–Ab–C
Bbm (vi)	Bb–Db–F	Db–F–Bb	F–Bb–Db

Progressions in This Key

Progression	Roman Numerals	Chords
I – IV – V	I – IV – V	Db – Gb – Ab
Pop Progression	I – V – vi – IV	Db – Ab – Bbm – Gb
50s Progression	I – vi – IV – V	Db – Bbm – Gb – Ab
Jazz ii–V–I	ii7 – V7 – Imaj7	Ebm7 – Ab7 – Dbmaj7
12-Bar Blues (I7-IV7-V7)	I7 – IV7 – V7	Db7 – Gb7 – Ab7

11.9 Key of A♭ Major

Scale

The A♭ Major scale: Ab – Bb – C – Db – Eb – F – G – Ab.

Chord Family (Diatonic Chords)

Numeral	Chord	Notes	Quality
I	Ab	Ab – C – Eb	Major
ii	Bbm	Bb – Db – F	minor
iii	Cm	C – Eb – G	minor
IV	Db	Db – F – Ab	Major
V	Eb	Eb – G – Bb	Major
vi	Fm	F – Ab – C	minor
vii°	Gdim	G – Bb – Db	Diminished

Triads to Drill (Root Position)

Chord	Notes (Root Position)
Ab (I)	Ab – C – Eb
Bbm (ii)	Bb – Db – F
Cm (iii)	C – Eb – G
Db (IV)	Db – F – Ab
Eb (V)	Eb – G – Bb
Fm (vi)	F – Ab – C
Gdim (vii°)	G – Bb – Db

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
Ab (I)	Ab–C–Eb	C–Eb–Ab	Eb–Ab–C
Db (IV)	Db–F–Ab	F–Ab–Db	Ab–Db–F
Eb (V)	Eb–G–Bb	G–Bb–Eb	Bb–Eb–G
Fm (vi)	F–Ab–C	Ab–C–F	C–F–Ab

Progressions in This Key

Progression	Roman Numerals	Chords
I – IV – V	I – IV – V	Ab – Db – Eb
Pop Progression	I – V – vi – IV	Ab – Eb – Fm – Db
50s Progression	I – vi – IV – V	Ab – Fm – Db – Eb
Jazz ii–V–I	ii7 – V7 – Imaj7	Bbm7 – Eb7 – Abmaj7
12-Bar Blues (I7-IV7-V7)	I7 – IV7 – V7	Ab7 – Db7 – Eb7

11.10 Key of E♭ Major

Scale

The E♭ Major scale: Eb – F – G – Ab – Bb – C – D – Eb.

Chord Family (Diatonic Chords)

Numeral	Chord	Notes	Quality
I	Eb	Eb – G – Bb	Major
ii	Fm	F – Ab – C	minor
iii	Gm	G – Bb – D	minor
IV	Ab	Ab – C – Eb	Major
V	Bb	Bb – D – F	Major
vi	Cm	C – Eb – G	minor
vii°	Ddim	D – F – Ab	Diminished

Triads to Drill (Root Position)

Chord	Notes (Root Position)
Eb (I)	Eb – G – Bb
Fm (ii)	F – Ab – C
Gm (iii)	G – Bb – D
Ab (IV)	Ab – C – Eb
Bb (V)	Bb – D – F
Cm (vi)	C – Eb – G
Ddim (vii°)	D – F – Ab

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
Eb (I)	Eb–G–Bb	G–Bb–Eb	Bb–Eb–G
Ab (IV)	Ab–C–Eb	C–Eb–Ab	Eb–Ab–C
Bb (V)	Bb–D–F	D–F–Bb	F–Bb–D
Cm (vi)	C–Eb–G	Eb–G–C	G–C–Eb

Progressions in This Key

Progression	Roman Numerals	Chords
I – IV – V	I – IV – V	Eb – Ab – Bb
Pop Progression	I – V – vi – IV	Eb – Bb – Cm – Ab
50s Progression	I – vi – IV – V	Eb – Cm – Ab – Bb
Jazz ii–V–I	ii7 – V7 – Imaj7	Fm7 – Bb7 – Ebmaj7
12-Bar Blues (I7-IV7-V7)	I7 – IV7 – V7	Eb7 – Ab7 – Bb7

11.11 Key of B♭ Major

Scale

The B♭ Major scale: Bb – C – D – Eb – F – G – A – Bb.

Chord Family (Diatonic Chords)

Numeral	Chord	Notes	Quality
I	Bb	Bb – D – F	Major
ii	Cm	C – Eb – G	minor
iii	Dm	D – F – A	minor
IV	Eb	Eb – G – Bb	Major
V	F	F – A – C	Major
vi	Gm	G – Bb – D	minor
vii°	Adim	A – C – Eb	Diminished

Triads to Drill (Root Position)

Chord	Notes (Root Position)
Bb (I)	Bb – D – F
Cm (ii)	C – Eb – G

Chord	Notes (Root Position)
Dm (iii)	D – F – A
Eb (IV)	Eb – G – Bb
F (V)	F – A – C
Gm (vi)	G – Bb – D
Adim (vii°)	A – C – Eb

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
Bb (I)	Bb–D–F	D–F–Bb	F–Bb–D
Eb (IV)	Eb–G–Bb	G–Bb–Eb	Bb–Eb–G
F (V)	F–A–C	A–C–F	C–F–A
Gm (vi)	G–Bb–D	Bb–D–G	D–G–Bb

Progressions in This Key

Progression	Roman Numerals	Chords
I – IV – V	I – IV – V	Bb – Eb – F
Pop Progression	I – V – vi – IV	Bb – F – Gm – Eb
50s Progression	I – vi – IV – V	Bb – Gm – Eb – F
Jazz ii–V–I	ii7 – V7 – Imaj7	Cm7 – F7 – Bbmaj7
12-Bar Blues (I7-IV7-V7)	I7 – IV7 – V7	Bb7 – Eb7 – F7

11.12 Key of F Major

Scale

The F Major scale: F – G – A – Bb – C – D – E – F.

Chord Family (Diatonic Chords)

Numeral	Chord	Notes	Quality
I	F	F – A – C	Major
ii	Gm	G – Bb – D	minor
iii	Am	A – C – E	minor
IV	Bb	Bb – D – F	Major
V	C	C – E – G	Major

Numeral	Chord	Notes	Quality
vi	Dm	D – F – A	minor
vii°	Edim	E – G – Bb	Diminished

Triads to Drill (Root Position)

Chord	Notes (Root Position)
F (I)	F – A – C
Gm (ii)	G – Bb – D
Am (iii)	A – C – E
Bb (IV)	Bb – D – F
C (V)	C – E – G
Dm (vi)	D – F – A
Edim (vii°)	E – G – Bb

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
F (I)	F–A–C	A–C–F	C–F–A
Bb (IV)	Bb–D–F	D–F–Bb	F–Bb–D
C (V)	C–E–G	E–G–C	G–C–E
Dm (vi)	D–F–A	F–A–D	A–D–F

Progressions in This Key

Progression	Roman Numerals	Chords
I – IV – V	I – IV – V	F – Bb – C
Pop Progression	I – V – vi – IV	F – C – Dm – Bb
50s Progression	I – vi – IV – V	F – Dm – Bb – C
Jazz ii–V–I	ii7 – V7 – Imaj7	Gm7 – C7 – Fmaj7
12-Bar Blues (I7-IV7-V7)	I7 – IV7 – V7	F7 – Bb7 – C7

All 12 Major Keys Complete

The major-key portion of this chapter now covers all 12 major keys — C, G, D, A, E, B, F#, Db, Ab, Eb, Bb, and F — each following the same five-step structure: Scale → Chord Family → Triads → Inversions → Progressions. The chapter now continues with the relative minor of each key.

11.13 Key of A minor (relative minor of C Major)

Scale (Natural Minor)

The A minor natural minor scale: A – B – C – D – E – F – G – A. This is the relative minor of C Major — the same seven notes, starting from the 6th degree.

Chord Family (Diatonic Chords, Natural Minor)

Numeral	Chord	Notes	Quality
i	Am	A – C – E	minor
ii°	Bdim	B – D – F	Diminished
III	C	C – E – G	Major
iv	Dm	D – F – A	minor
v	Em	E – G – B	minor
VI	F	F – A – C	Major
VII	G	G – B – D	Major

Harmonic Minor & the V Chord

In natural minor, the v chord is minor and has little pull toward "home." Raising the 7th scale degree (creating the harmonic minor scale) turns v into a Major V chord — E Major (E – G# – B) — giving minor keys the same strong V→i resolution that major keys get from their V→I.

Triads to Drill (Root Position)

Chord	Notes (Root Position)
Am (i)	A – C – E
Bdim (ii°)	B – D – F
C (III)	C – E – G
Dm (iv)	D – F – A
Em (v)	E – G – B
F (VI)	F – A – C
G (VII)	G – B – D

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
Am (i)	A–C–E	C–E–A	E–A–C
Dm (iv)	D–F–A	F–A–D	A–D–F
E (V)	E–G#–B	G#–B–E	B–E–G#

Chord	Root Position	1st Inversion	2nd Inversion
F (VI)	F–A–C	A–C–F	C–F–A

Progressions in This Key

Progression	Roman Numerals	Chords
Natural Minor Cadence	i – iv – v	Am – Dm – Em
Harmonic Minor Cadence	i – iv – V	Am – Dm – E
Pop/Rock Minor	i – VI – III – VII	Am – F – C – G
Andalusian Cadence	i – VII – VI – V	Am – G – F – E

11.14 Key of E minor (relative minor of G Major)

Scale (Natural Minor)

The E minor natural minor scale: E – F# – G – A – B – C – D – E. This is the relative minor of G Major — the same seven notes, starting from the 6th degree.

Chord Family (Diatonic Chords, Natural Minor)

Numeral	Chord	Notes	Quality
i	Em	E – G – B	minor
ii°	F#dim	F# – A – C	Diminished
III	G	G – B – D	Major
iv	Am	A – C – E	minor
v	Bm	B – D – F#	minor
VI	C	C – E – G	Major
VII	D	D – F# – A	Major

> **Harmonic Minor & the V Chord**
> In natural minor, the v chord is minor and has little pull toward "home." Raising the 7th scale degree (creating the harmonic minor scale) turns v into a Major V chord — B Major (B – D# – F#) — giving minor keys the same strong V→i resolution that major keys get from their V→I.

Triads to Drill (Root Position)

Chord	Notes (Root Position)
Em (i)	E – G – B
F#dim (ii°)	F# – A – C

Chord	Notes (Root Position)
G (III)	G – B – D
Am (iv)	A – C – E
Bm (v)	B – D – F#
C (VI)	C – E – G
D (VII)	D – F# – A

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
Em (i)	E–G–B	G–B–E	B–E–G
Am (iv)	A–C–E	C–E–A	E–A–C
B (V)	B–D#–F#	D#–F#–B	F#–B–D#
C (VI)	C–E–G	E–G–C	G–C–E

Progressions in This Key

Progression	Roman Numerals	Chords
Natural Minor Cadence	i – iv – v	Em – Am – Bm
Harmonic Minor Cadence	i – iv – V	Em – Am – B
Pop/Rock Minor	i – VI – III – VII	Em – C – G – D
Andalusian Cadence	i – VII – VI – V	Em – D – C – B

11.15 Key of B minor (relative minor of D Major)

Scale (Natural Minor)

The B minor natural minor scale: B – C# – D – E – F# – G – A – B. This is the relative minor of D Major — the same seven notes, starting from the 6th degree.

Chord Family (Diatonic Chords, Natural Minor)

Numeral	Chord	Notes	Quality
i	Bm	B – D – F#	minor
ii°	C#dim	C# – E – G	Diminished
III	D	D – F# – A	Major
iv	Em	E – G – B	minor
v	F#m	F# – A – C#	minor

Numeral	Chord	Notes	Quality
VI	G	G – B – D	Major
VII	A	A – C# – E	Major

Harmonic Minor & the V Chord

In natural minor, the v chord is minor and has little pull toward "home." Raising the 7th scale degree (creating the harmonic minor scale) turns v into a Major V chord — F# Major (F# – A# – C#) — giving minor keys the same strong V→i resolution that major keys get from their V→I.

Triads to Drill (Root Position)

Chord	Notes (Root Position)
Bm (i)	B – D – F#
C#dim (ii°)	C# – E – G
D (III)	D – F# – A
Em (iv)	E – G – B
F#m (v)	F# – A – C#
G (VI)	G – B – D
A (VII)	A – C# – E

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
Bm (i)	B–D–F#	D–F#–B	F#–B–D
Em (iv)	E–G–B	G–B–E	B–E–G
F# (V)	F#–A#–C#	A#–C#–F#	C#–F#–A#
G (VI)	G–B–D	B–D–G	D–G–B

Progressions in This Key

Progression	Roman Numerals	Chords
Natural Minor Cadence	i – iv – v	Bm – Em – F#m
Harmonic Minor Cadence	i – iv – V	Bm – Em – F#
Pop/Rock Minor	i – VI – III – VII	Bm – G – D – A
Andalusian Cadence	i – VII – VI – V	Bm – A – G – F#

11.16 Key of F# minor (relative minor of A Major)

Scale (Natural Minor)

The F# minor natural minor scale: F# – G# – A – B – C# – D – E – F#. This is the relative minor of A Major — the same seven notes, starting from the 6th degree.

Chord Family (Diatonic Chords, Natural Minor)

Numeral	Chord	Notes	Quality
i	F#m	F# – A – C#	minor
ii°	G#dim	G# – B – D	Diminished
III	A	A – C# – E	Major
iv	Bm	B – D – F#	minor
v	C#m	C# – E – G#	minor
VI	D	D – F# – A	Major
VII	E	E – G# – B	Major

> **Harmonic Minor & the V Chord**
> In natural minor, the v chord is minor and has little pull toward "home." Raising the 7th scale degree (creating the harmonic minor scale) turns v into a Major V chord — C# Major (C# – E# – G#) — giving minor keys the same strong V→i resolution that major keys get from their V→I.

Triads to Drill (Root Position)

Chord	Notes (Root Position)
F#m (i)	F# – A – C#
G#dim (ii°)	G# – B – D
A (III)	A – C# – E
Bm (iv)	B – D – F#
C#m (v)	C# – E – G#
D (VI)	D – F# – A
E (VII)	E – G# – B

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
F#m (i)	F#–A–C#	A–C#–F#	C#–F#–A
Bm (iv)	B–D–F#	D–F#–B	F#–B–D
C# (V)	C#–E#–G#	E#–G#–C#	G#–C#–E#

Chord	Root Position	1st Inversion	2nd Inversion
D (VI)	D–F#–A	F#–A–D	A–D–F#

Progressions in This Key

Progression	Roman Numerals	Chords
Natural Minor Cadence	i – iv – v	F#m – Bm – C#m
Harmonic Minor Cadence	i – iv – V	F#m – Bm – C#
Pop/Rock Minor	i – VI – III – VII	F#m – D – A – E
Andalusian Cadence	i – VII – VI – V	F#m – E – D – C#

11.17 Key of C# minor (relative minor of E Major)

Scale (Natural Minor)

The C# minor natural minor scale: C# – D# – E – F# – G# – A – B – C#. This is the relative minor of E Major — the same seven notes, starting from the 6th degree.

Chord Family (Diatonic Chords, Natural Minor)

Numeral	Chord	Notes	Quality
i	C#m	C# – E – G#	minor
ii°	D#dim	D# – F# – A	Diminished
III	E	E – G# – B	Major
iv	F#m	F# – A – C#	minor
v	G#m	G# – B – D#	minor
VI	A	A – C# – E	Major
VII	B	B – D# – F#	Major

> **Harmonic Minor & the V Chord**
> In natural minor, the v chord is minor and has little pull toward "home." Raising the 7th scale degree (creating the harmonic minor scale) turns v into a Major V chord — G# Major (G# – B# – D#) — giving minor keys the same strong V→i resolution that major keys get from their V→I.

Triads to Drill (Root Position)

Chord	Notes (Root Position)
C#m (i)	C# – E – G#
D#dim (ii°)	D# – F# – A

Chord	Notes (Root Position)
E (III)	E – G# – B
F#m (iv)	F# – A – C#
G#m (v)	G# – B – D#
A (VI)	A – C# – E
B (VII)	B – D# – F#

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
C#m (i)	C#–E–G#	E–G#–C#	G#–C#–E
F#m (iv)	F#–A–C#	A–C#–F#	C#–F#–A
G# (V)	G#–B#–D#	B#–D#–G#	D#–G#–B#
A (VI)	A–C#–E	C#–E–A	E–A–C#

Progressions in This Key

Progression	Roman Numerals	Chords
Natural Minor Cadence	i – iv – v	C#m – F#m – G#m
Harmonic Minor Cadence	i – iv – V	C#m – F#m – G#
Pop/Rock Minor	i – VI – III – VII	C#m – A – E – B
Andalusian Cadence	i – VII – VI – V	C#m – B – A – G#

11.18 Key of G# minor (relative minor of B Major)

Scale (Natural Minor)

The G# minor natural minor scale: G# – A# – B – C# – D# – E – F# – G#. This is the relative minor of B Major — the same seven notes, starting from the 6th degree.

Chord Family (Diatonic Chords, Natural Minor)

Numeral	Chord	Notes	Quality
i	G#m	G# – B – D#	minor
ii°	A#dim	A# – C# – E	Diminished
III	B	B – D# – F#	Major
iv	C#m	C# – E – G#	minor
v	D#m	D# – F# – A#	minor

Numeral	Chord	Notes	Quality
VI	E	E – G# – B	Major
VII	F#	F# – A# – C#	Major

Harmonic Minor & the V Chord

In natural minor, the v chord is minor and has little pull toward "home." Raising the 7th scale degree (creating the harmonic minor scale) turns v into a Major V chord — D# Major (spelled D#–G–A#; strict theory: D#–Fx–A#) (D# – G – A#) — giving minor keys the same strong V→i resolution that major keys get from their V→I.

Triads to Drill (Root Position)

Chord	Notes (Root Position)
G#m (i)	G# – B – D#
A#dim (ii°)	A# – C# – E
B (III)	B – D# – F#
C#m (iv)	C# – E – G#
D#m (v)	D# – F# – A#
E (VI)	E – G# – B
F# (VII)	F# – A# – C#

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
G#m (i)	G#–B–D#	B–D#–G#	D#–G#–B
C#m (iv)	C#–E–G#	E–G#–C#	G#–C#–E
D# (V)	D#–G–A#	G–A#–D#	A#–D#–G
E (VI)	E–G#–B	G#–B–E	B–E–G#

Progressions in This Key

Progression	Roman Numerals	Chords
Natural Minor Cadence	i – iv – v	G#m – C#m – D#m
Harmonic Minor Cadence	i – iv – V	G#m – C#m – D#
Pop/Rock Minor	i – VI – III – VII	G#m – E – B – F#
Andalusian Cadence	i – VII – VI – V	G#m – F# – E – D#

11.19 Key of D# minor (relative minor of F# Major)

Scale (Natural Minor)

The D# minor natural minor scale: D# – E# – F# – G# – A# – B – C# – D#. This is the relative minor of F# Major — the same seven notes, starting from the 6th degree.

Chord Family (Diatonic Chords, Natural Minor)

Numeral	Chord	Notes	Quality
i	D#m	D# – F# – A#	minor
ii°	E#dim	E# – G# – B	Diminished
III	F#	F# – A# – C#	Major
iv	G#m	G# – B – D#	minor
v	A#m	A# – C# – E#	minor
VI	B	B – D# – F#	Major
VII	C#	C# – E# – G#	Major

> **Harmonic Minor & the V Chord**
> In natural minor, the v chord is minor and has little pull toward "home." Raising the 7th scale degree (creating the harmonic minor scale) turns v into a Major V chord — A# Major (spelled A#–C##–E#; commonly played as Bb Major: Bb–D–F) (A# – D – F (practical spelling)) — giving minor keys the same strong V→i resolution that major keys get from their V→I.

Triads to Drill (Root Position)

Chord	Notes (Root Position)
D#m (i)	D# – F# – A#
E#dim (ii°)	E# – G# – B
F# (III)	F# – A# – C#
G#m (iv)	G# – B – D#
A#m (v)	A# – C# – E#
B (VI)	B – D# – F#
C# (VII)	C# – E# – G#

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
D#m (i)	D#–F#–A#	F#–A#–D#	A#–D#–F#
G#m (iv)	G#–B–D#	B–D#–G#	D#–G#–B

Chord	Root Position	1st Inversion	2nd Inversion
A#/Bb (V)	Bb–D–F	D–F–Bb	F–Bb–D
B (VI)	B–D#–F#	D#–F#–B	F#–B–D#

Progressions in This Key

Progression	Roman Numerals	Chords
Natural Minor Cadence	i – iv – v	D#m – G#m – A#m
Harmonic Minor Cadence	i – iv – V	D#m – G#m – A#(Bb)
Pop/Rock Minor	i – VI – III – VII	D#m – B – F# – C#
Andalusian Cadence	i – VII – VI – V	D#m – C# – B – A#(Bb)

11.20 Key of B♭ minor (relative minor of D♭ Major)

Scale (Natural Minor)

The B♭ minor natural minor scale: Bb – C – Db – Eb – F – Gb – Ab – Bb. This is the relative minor of D♭ Major — the same seven notes, starting from the 6th degree.

Chord Family (Diatonic Chords, Natural Minor)

Numeral	Chord	Notes	Quality
i	Bbm	Bb – Db – F	minor
ii°	Cdim	C – Eb – Gb	Diminished
III	Db	Db – F – Ab	Major
iv	Ebm	Eb – Gb – Bb	minor
v	Fm	F – Ab – C	minor
VI	Gb	Gb – Bb – Db	Major
VII	Ab	Ab – C – Eb	Major

> **Harmonic Minor & the V Chord**
> In natural minor, the v chord is minor and has little pull toward "home." Raising the 7th scale degree (creating the harmonic minor scale) turns v into a Major V chord — F Major (F – A – C) — giving minor keys the same strong V→i resolution that major keys get from their V→I.

Triads to Drill (Root Position)

Chord	Notes (Root Position)
Bbm (i)	Bb – Db – F

Chord	Notes (Root Position)
Cdim (ii°)	C – Eb – Gb
Db (III)	Db – F – Ab
Ebm (iv)	Eb – Gb – Bb
Fm (v)	F – Ab – C
Gb (VI)	Gb – Bb – Db
Ab (VII)	Ab – C – Eb

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
Bbm (i)	Bb–Db–F	Db–F–Bb	F–Bb–Db
Ebm (iv)	Eb–Gb–Bb	Gb–Bb–Eb	Bb–Eb–Gb
F (V)	F–A–C	A–C–F	C–F–A
Gb (VI)	Gb–Bb–Db	Bb–Db–Gb	Db–Gb–Bb

Progressions in This Key

Progression	Roman Numerals	Chords
Natural Minor Cadence	i – iv – v	Bbm – Ebm – Fm
Harmonic Minor Cadence	i – iv – V	Bbm – Ebm – F
Pop/Rock Minor	i – VI – III – VII	Bbm – Gb – Db – Ab
Andalusian Cadence	i – VII – VI – V	Bbm – Ab – Gb – F

11.21 Key of F minor (relative minor of A♭ Major)

Scale (Natural Minor)

The F minor natural minor scale: F – G – Ab – Bb – C – Db – Eb – F. This is the relative minor of A♭ Major — the same seven notes, starting from the 6th degree.

Chord Family (Diatonic Chords, Natural Minor)

Numeral	Chord	Notes	Quality
i	Fm	F – Ab – C	minor
ii°	Gdim	G – Bb – Db	Diminished
III	Ab	Ab – C – Eb	Major
iv	Bbm	Bb – Db – F	minor

Numeral	Chord	Notes	Quality
v	Cm	C – Eb – G	minor
VI	Db	Db – F – Ab	Major
VII	Eb	Eb – G – Bb	Major

Harmonic Minor & the V Chord

In natural minor, the v chord is minor and has little pull toward "home." Raising the 7th scale degree (creating the harmonic minor scale) turns v into a Major V chord — C Major (C – E – G) — giving minor keys the same strong V→i resolution that major keys get from their V→I.

Triads to Drill (Root Position)

Chord	Notes (Root Position)
Fm (i)	F – Ab – C
Gdim (ii°)	G – Bb – Db
Ab (III)	Ab – C – Eb
Bbm (iv)	Bb – Db – F
Cm (v)	C – Eb – G
Db (VI)	Db – F – Ab
Eb (VII)	Eb – G – Bb

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
Fm (i)	F–Ab–C	Ab–C–F	C–F–Ab
Bbm (iv)	Bb–Db–F	Db–F–Bb	F–Bb–Db
C (V)	C–E–G	E–G–C	G–C–E
Db (VI)	Db–F–Ab	F–Ab–Db	Ab–Db–F

Progressions in This Key

Progression	Roman Numerals	Chords
Natural Minor Cadence	i – iv – v	Fm – Bbm – Cm
Harmonic Minor Cadence	i – iv – V	Fm – Bbm – C
Pop/Rock Minor	i – VI – III – VII	Fm – Db – Ab – Eb
Andalusian Cadence	i – VII – VI – V	Fm – Eb – Db – C

11.22 Key of C minor (relative minor of E♭ Major)

Scale (Natural Minor)

The C minor natural minor scale: C – D – Eb – F – G – Ab – Bb – C. This is the relative minor of E♭ Major — the same seven notes, starting from the 6th degree.

Chord Family (Diatonic Chords, Natural Minor)

Numeral	Chord	Notes	Quality
i	Cm	C – Eb – G	minor
ii°	Ddim	D – F – Ab	Diminished
III	Eb	Eb – G – Bb	Major
iv	Fm	F – Ab – C	minor
v	Gm	G – Bb – D	minor
VI	Ab	Ab – C – Eb	Major
VII	Bb	Bb – D – F	Major

> **Harmonic Minor & the V Chord**
> In natural minor, the v chord is minor and has little pull toward "home." Raising the 7th scale degree (creating the harmonic minor scale) turns v into a Major V chord — G Major (G – B – D) — giving minor keys the same strong V→i resolution that major keys get from their V→I.

Triads to Drill (Root Position)

Chord	Notes (Root Position)
Cm (i)	C – Eb – G
Ddim (ii°)	D – F – Ab
Eb (III)	Eb – G – Bb
Fm (iv)	F – Ab – C
Gm (v)	G – Bb – D
Ab (VI)	Ab – C – Eb
Bb (VII)	Bb – D – F

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
Cm (i)	C–Eb–G	Eb–G–C	G–C–Eb
Fm (iv)	F–Ab–C	Ab–C–F	C–F–Ab
G (V)	G–B–D	B–D–G	D–G–B

Chord	Root Position	1st Inversion	2nd Inversion
Ab (VI)	Ab–C–Eb	C–Eb–Ab	Eb–Ab–C

Progressions in This Key

Progression	Roman Numerals	Chords
Natural Minor Cadence	i – iv – v	Cm – Fm – Gm
Harmonic Minor Cadence	i – iv – V	Cm – Fm – G
Pop/Rock Minor	i – VI – III – VII	Cm – Ab – Eb – Bb
Andalusian Cadence	i – VII – VI – V	Cm – Bb – Ab – G

11.23 Key of G minor (relative minor of B♭ Major)

Scale (Natural Minor)

The G minor natural minor scale: G – A – Bb – C – D – Eb – F – G. This is the relative minor of B♭ Major — the same seven notes, starting from the 6th degree.

Chord Family (Diatonic Chords, Natural Minor)

Numeral	Chord	Notes	Quality
i	Gm	G – Bb – D	minor
ii°	Adim	A – C – Eb	Diminished
III	Bb	Bb – D – F	Major
iv	Cm	C – Eb – G	minor
v	Dm	D – F – A	minor
VI	Eb	Eb – G – Bb	Major
VII	F	F – A – C	Major

> **Harmonic Minor & the V Chord**
> In natural minor, the v chord is minor and has little pull toward "home." Raising the 7th scale degree (creating the harmonic minor scale) turns v into a Major V chord — D Major (D – F# – A) — giving minor keys the same strong V→i resolution that major keys get from their V→I.

Triads to Drill (Root Position)

Chord	Notes (Root Position)
Gm (i)	G – Bb – D
Adim (ii°)	A – C – Eb

Chord	Notes (Root Position)
Bb (III)	Bb – D – F
Cm (iv)	C – Eb – G
Dm (v)	D – F – A
Eb (VI)	Eb – G – Bb
F (VII)	F – A – C

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
Gm (i)	G–Bb–D	Bb–D–G	D–G–Bb
Cm (iv)	C–Eb–G	Eb–G–C	G–C–Eb
D (V)	D–F#–A	F#–A–D	A–D–F#
Eb (VI)	Eb–G–Bb	G–Bb–Eb	Bb–Eb–G

Progressions in This Key

Progression	Roman Numerals	Chords
Natural Minor Cadence	i – iv – v	Gm – Cm – Dm
Harmonic Minor Cadence	i – iv – V	Gm – Cm – D
Pop/Rock Minor	i – VI – III – VII	Gm – Eb – Bb – F
Andalusian Cadence	i – VII – VI – V	Gm – F – Eb – D

11.24 Key of D minor (relative minor of F Major)

Scale (Natural Minor)

The D minor natural minor scale: D – E – F – G – A – Bb – C – D. This is the relative minor of F Major — the same seven notes, starting from the 6th degree.

Chord Family (Diatonic Chords, Natural Minor)

Numeral	Chord	Notes	Quality
i	Dm	D – F – A	minor
ii°	Edim	E – G – Bb	Diminished
III	F	F – A – C	Major
iv	Gm	G – Bb – D	minor
v	Am	A – C – E	minor

Numeral	Chord	Notes	Quality
VI	Bb	Bb – D – F	Major
VII	C	C – E – G	Major

Harmonic Minor & the V Chord

In natural minor, the v chord is minor and has little pull toward "home." Raising the 7th scale degree (creating the harmonic minor scale) turns v into a Major V chord — A Major (A – C# – E) — giving minor keys the same strong V→i resolution that major keys get from their V→I.

Triads to Drill (Root Position)

Chord	Notes (Root Position)
Dm (i)	D – F – A
Edim (ii°)	E – G – Bb
F (III)	F – A – C
Gm (iv)	G – Bb – D
Am (v)	A – C – E
Bb (VI)	Bb – D – F
C (VII)	C – E – G

Inversions Practice

Chord	Root Position	1st Inversion	2nd Inversion
Dm (i)	D–F–A	F–A–D	A–D–F
Gm (iv)	G–Bb–D	Bb–D–G	D–G–Bb
A (V)	A–C#–E	C#–E–A	E–A–C#
Bb (VI)	Bb–D–F	D–F–Bb	F–Bb–D

Progressions in This Key

Progression	Roman Numerals	Chords
Natural Minor Cadence	i – iv – v	Dm – Gm – Am
Harmonic Minor Cadence	i – iv – V	Dm – Gm – A
Pop/Rock Minor	i – VI – III – VII	Dm – Bb – F – C
Andalusian Cadence	i – VII – VI – V	Dm – C – Bb – A

Chapter Complete — What's Next

Chapter 11 now covers all 12 major keys plus all 12 relative minors — a complete reference spanning every major and minor key. Chapter 12 pulls everything from this book together into a final set of comprehensive review exercises.

Chapter 12 · 25 Exercises for Students

This final chapter is a comprehensive review — twenty-five exercises that pull together everything from Chapters 1 through 11 into one capstone practice set. The first fifteen are ordered roughly by difficulty, moving from foundational recall through full-key routines to an original composition; the final ten push further into real-world application — playing in unfamiliar keys on request, arranging by ear, and teaching the material back to someone else. Use this chapter as a benchmark: work through it now, then return to it periodically to check how much faster and more confident you have become.

> **How to Use This Chapter**
> Do not rush through all fifteen in one sitting the first time. Spread them across several practice sessions, and always use a real keyboard and a metronome where indicated. If any exercise feels shaky, return to the chapter referenced in parentheses before continuing.

Comprehensive Review Exercises

Exercise 1: Formula Recall Sprint (Chapters 1, 3–5)

- Pick 4 random root notes. For each root, build the Major (1-3-5), minor (1-b3-5), Diminished (1-b3-b5), and Augmented (1-3-#5) triads.
- Say each formula out loud before playing it — do not rely on muscle memory alone.
- Time yourself: aim for all 16 triads (4 roots × 4 qualities) in under 3 minutes.

Self-check:

☐ I built all 16 triads correctly.

☐ I finished within 3 minutes without checking a reference table.

Exercise 2: Speed Harmonization Challenge (Chapters 2, 7)

- Pick a key you have not practiced recently. Spell its Major scale, then find all seven diatonic chords (I–vii°) using the fast method.
- Label each chord with its Roman numeral and quality as you go.
- Repeat in a second key, aiming for under 20 seconds each.

Self-check:

☐ I correctly harmonized both keys.

☐ My second key took under 20 seconds.

Exercise 3: All 12 Major Triads Marathon (Chapter 3)

- Play all 12 Major triads in root position, in circle-of-fifths order, without stopping.
- Say each chord's three note names out loud as you play it.
- Repeat immediately a second time, slightly faster than the first pass.

Self-check:

☐ I completed both passes without consulting a reference table.

☐ My second pass was noticeably faster than my first.

Exercise 4: All 12 Minor Triads Marathon (Chapter 4)

- Play all 12 minor triads in root position, in circle-of-fifths order, without stopping.
- For 4 of the 12, convert immediately to the Major triad on the same root, then back to minor.
- Say each chord's quality out loud as you play it.

Self-check:

☐ I completed the full minor circle without stopping.

☐ My 4 Major/minor conversions were accurate and immediate.

Exercise 5: Four-Quality Same-Root Test (Chapter 5)

- Pick 3 random roots. For each, play Major, minor, Diminished, and Augmented in a row.
- Identify out loud exactly which note changes at each step (3rd, then 5th, then 5th again).
- Have someone quiz you by naming a quality at random; play it immediately on a root they choose.

Self-check:

☐ I played all 12 chords (3 roots × 4 qualities) correctly.

☐ I responded correctly to at least 4 out of 5 random quiz prompts.

Exercise 6: Voice-Leading Challenge (Chapter 6)

- Choose any 4-chord progression from Chapter 9 in a key of your choice.
- Find the smoothest possible inversion path through all 4 chords using minimal hand movement.
- Play the progression through 3 times, evaluating your own smoothness each time.

Self-check:

☐ I found an inversion path that clearly reduces hand movement compared to root position only.

☐ My third playthrough was smoother than my first.

Exercise 7: Chord Family & Roman Numeral Quiz (Chapter 7)

- Without looking at any reference, sort I, ii, iii, IV, V, vi, and vii° into Tonic, Subdominant, and Dominant families.
- In a key of your choice, name and play the I, IV, and V chords within 10 seconds.
- Explain out loud, in your own words, why I and vi can often substitute for each other.

Self-check:

☐ I sorted at least 6 of 7 numerals into the correct family.

☐ I named and played I, IV, V within 10 seconds.

Exercise 8: Seventh Chord Build-and-Identify (Chapter 8)

- On a single root of your choice, build all five 7th-chord qualities: maj7, 7, m7, m7♭5, dim7.
- Have someone play one at random (or use a recording/app); identify which quality you heard.
- Play a ii7–V7–Imaj7 progression in 2 different keys.

Self-check:

☐ I built all 5 qualities correctly on my chosen root.

☐ I correctly identified the quality at least 4 times out of 5 random plays.

Exercise 9: Progression Playback (Chapter 9)

- Choose 3 of the 5 core progressions from Chapter 9 (I-IV-V, I-V-vi-IV, ii-V-I, I-vi-IV-V, or the 12-bar blues).

- Play each one in 2 different keys, for a total of 6 performances.
- For at least one, recall and name a real song that uses that progression.

Self-check:

☐ I performed all 6 progression-key combinations correctly.

☐ I identified at least one real song matching one of the progressions.

Exercise 10: Metronome & Rhythm Pattern Test (Chapter 10)

- Set a metronome to 80 BPM. Play any progression from Chapter 9 using block chords, staying locked to the click.
- Repeat the same progression using quarter-note pulse, then broken chords, then syncopated comping.
- Add a simple left-hand root-note pattern under your right-hand chords for at least one repetition.

Self-check:

☐ I stayed locked to the metronome across all 4 rhythm patterns.

☐ My left-hand/right-hand coordination held together for at least one full pass.

Exercise 11: Full Major-Key Routine (Chapter 11)

- Pick a random major key from Chapter 11 (roll a die, or ask someone to choose one of the 12).
- Work through the complete five-step routine for that key: Scale → Chord Family → Triads → Inversions → Progressions.
- Time the entire routine and note how long it takes from start to finish.

Self-check:

☐ I completed all 5 steps for my chosen key without consulting notes beyond Chapter 11 itself.

☐ I noted my completion time to compare against future attempts.

Exercise 12: Full Minor-Key Routine (Chapter 11)

- Pick a random relative minor key from Chapter 11.
- Work through its natural minor scale, chord family, triads, inversions, and progressions, including the harmonic-minor V chord.
- Play the Andalusian cadence (i–VII–VI–V) for that key at least twice.

Self-check:

☐ I completed the full minor-key routine for my chosen key.

☐ I played the Andalusian cadence correctly at least twice.

Exercise 13: Mixed Ear Training Marathon (Chapters 3–5, 8)

- Have someone (or a recording/app) play 15 chords in random order, drawn from Major, minor, Diminished, Augmented, and any 7th-chord quality.
- For each one, identify the quality as specifically as you can.
- Review every chord you missed by playing it again slowly next to its closest "neighbor" quality.

Self-check:

☐ I correctly identified at least 10 of the 15 chords.

☐ I reviewed and understood the reason behind each mistake.

Exercise 14: 12-Bar Blues Performance Test (Chapters 9–10)

- Set a metronome to a comfortable tempo. Perform the full 12-bar blues in a key of your choice, using dominant 7th chords throughout.
- Add a left-hand root-note pattern on beat 1 of each bar.
- Loop the full 12 bars at least 3 times without stopping or losing the beat.

Self-check:

☐ I performed the complete 12-bar blues with left-hand accompaniment.

☐ I looped it 3 times without stopping or losing tempo.

Exercise 15: Compose and Perform an Original Piece (Capstone)

- Choose any major or minor key from Chapter 11. Write an original 8-chord progression using only diatonic chords from that key (major or natural/harmonic minor).
- Label every chord with its Roman numeral and choose smooth-voice-leading inversions for the whole progression.
- Choose a rhythm pattern from Chapter 10 and perform your progression from start to finish, at tempo, at least twice through without stopping.

Self-check:

☐ I created an original 8-chord progression using correctly labeled diatonic chords.

☐ I performed it twice through, at tempo, without stopping.

Extended Practice Exercises

These ten additional exercises push beyond recall and into real-world application — playing from memory in unfamiliar keys, arranging by ear, and teaching the material back to someone else.

Exercise 16: Instant Transposition on Request

- Have someone call out a familiar progression by numerals only (e.g., "I-V-vi-IV") and a key at random.
- Play it immediately, without pausing to calculate on paper.
- Repeat for 5 different key/progression combinations, alternating major and minor keys.

Self-check:

☐ I played all 5 combinations correctly without writing anything down first.

☐ My response time got noticeably faster by the 5th combination.

Exercise 17: Left-Hand Bass Line Independence

- Choose any progression from Chapter 9. Play simple left-hand root notes on beats 1 and 3 of each bar.
- While the left hand holds that steady pattern, play the right-hand chords using a rhythm pattern from Chapter 10.
- Once comfortable, change the left hand to a walking bass line that moves stepwise between each chord's root.

Self-check:

☐ I maintained a steady left-hand pattern while the right hand played chords.

☐ My walking bass line connected smoothly between chord roots.

Exercise 18: Learn a Song by Ear

- Choose a simple song you know well. Listen closely and try to identify its chord progression using Roman numerals, not letter names, first.
- Once you have the numerals, work out the actual key and play the song's chords in that key.
- Play through the full song's chord progression at least twice without stopping.

Self-check:

☐ I correctly identified the progression using Roman numerals before naming actual chords.

☐ I played the full song's progression through twice without stopping.

Exercise 19: Diminished & Augmented as Passing Chords

- Take any I-IV-V progression in a key of your choice. Insert a passing diminished 7th chord between the I and the IV (e.g., C - C#dim7 - F).
- Try the same idea inserting an augmented triad as a passing chord between two chords a whole step apart.
- Play the progression with and without the passing chords, and compare how each version feels.

Self-check:

☐ I successfully inserted a passing diminished chord that resolves smoothly.

☐ I can describe, in my own words, how the passing chord changed the feel of the progression.

Exercise 20: Sight-Reading Chord Charts

- Write out (or ask someone to write) a simple 8-chord chart using only Roman numerals, no letter names, in a key you choose at the moment you see it.
- Read through and play the chart in real time, without stopping to work out each chord in advance.
- Repeat with a second chart in a different key, aiming to read it more fluently than the first.

Self-check:

☐ I played through the first chart with no more than 2 hesitations.

☐ My second chart was noticeably smoother than my first.

Exercise 21: Silent / Mental Practice

- Away from the keyboard, close your eyes and mentally "play" the diatonic chords of a key you choose, picturing the keys under your fingers.
- Mentally work through a full progression from Chapter 9 the same way, imagining the hand movement and sound.
- Return to the keyboard and play the same progression, checking how accurately your mental image matched reality.

Self-check:

☐ I could mentally picture at least 5 of the 7 diatonic chords accurately.

☐ My keyboard performance closely matched what I had mentally rehearsed.

Exercise 22: Tempo Ramp-Up Drill

- Set a metronome to a comfortable, slow tempo. Play a progression of your choice cleanly at that tempo, 4 times through.
- Increase the tempo by 10 BPM and repeat, without introducing errors.

- Continue increasing by 10 BPM increments until you notice mistakes creeping in, then back off to the last clean tempo.

Self-check:

☐ I increased tempo at least 3 times while staying accurate.

☐ I identified my current clean maximum tempo for this progression.

Exercise 23: Modulation Between Two Keys

- Choose two closely related keys (e.g., C Major and G Major, or C Major and its relative A minor).
- Play a short progression in the first key, then find a smooth chord (often the V of the new key, or a shared chord) to pivot into the second key.
- Play a short progression in the second key to complete the modulation, then try reversing the process back to the first key.

Self-check:

☐ I found a chord that convincingly bridges the two keys.

☐ I completed the modulation in both directions.

Exercise 24: Teach It Back

- Choose one concept from this book (e.g., building a Major triad, or the diatonic quality pattern) and explain it out loud as if teaching a beginner student, without looking at the book.
- Demonstrate the concept at the keyboard while you explain it.
- Ask yourself (or a listener) whether the explanation was clear enough for someone with no prior knowledge to follow.

Self-check:

☐ I explained the concept accurately without referring back to the book.

☐ My keyboard demonstration matched my spoken explanation.

Exercise 25: Full-Book Random Recall Challenge

- Write the titles of Chapters 1 through 11 on separate slips of paper (or list them) and draw one at random.
- Without any reference, perform the core skill from that chapter — e.g., if you draw Chapter 5, build and play all 4 triad qualities on 2 random roots.
- Repeat by drawing 3 more chapters at random, completing each chapter's core skill before moving on.

Self-check:

☐ I completed all 4 random chapter challenges without consulting notes.

☐ I can honestly self-rate my confidence in each of the 4 areas drawn.

Course Complete

Congratulations on working through all twelve chapters of this book — from the very first definition of a chord to composing and performing an original progression, plus twenty-five capstone exercises covering every angle of real-world keyboard practice. Return to any chapter or exercise whenever you want to sharpen a specific skill, and keep building new pieces of vocabulary onto this foundation.